Generalized Rank-based Evaluation for Knowledge Graph Completion : Challenges, Framework, and Analyses

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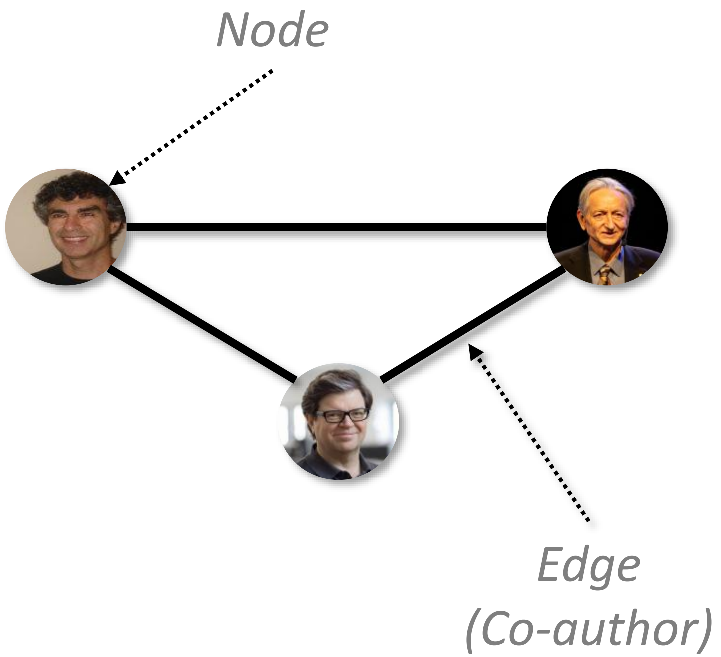
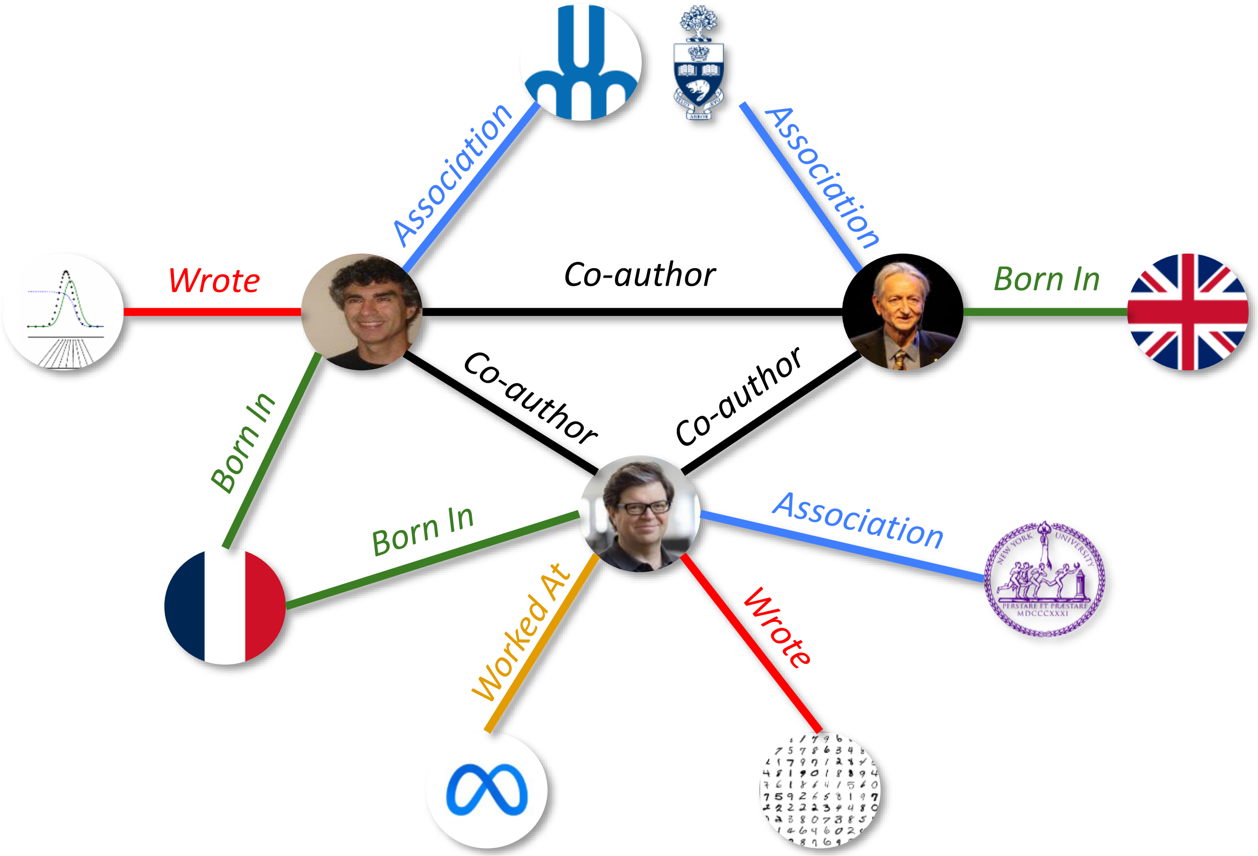
KNOWLEDGE GRAPH COMPLETION

■ What is a Knowledge Graph (KG)?

[Ordinary Graph]



[Knowledge Graph]

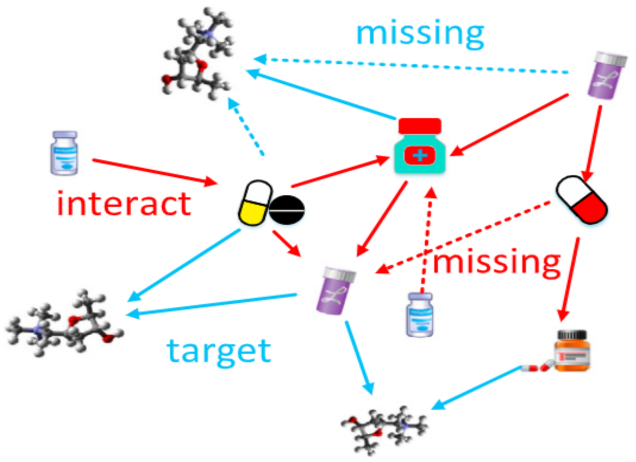


KNOWLEDGE GRAPH COMPLETION

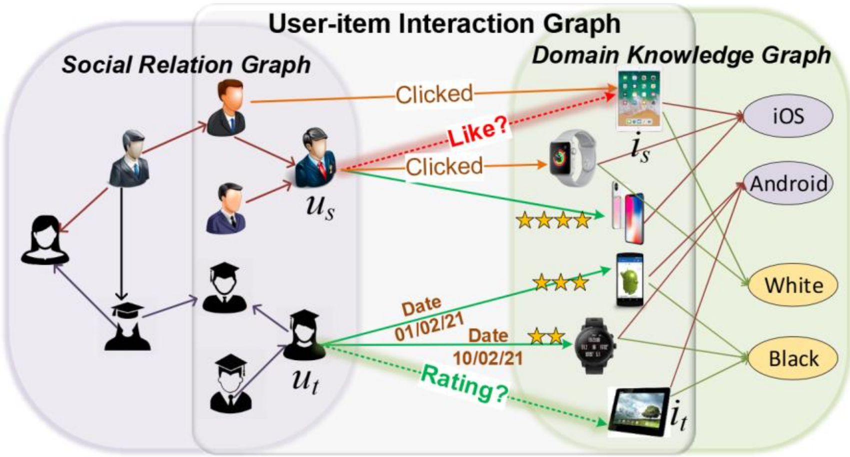
■ Multi-relationship Between Entities

- ☐ **Edge : Triple** (head entity, relation, tail entity)
- ☐ Rich representation → applied in various domains

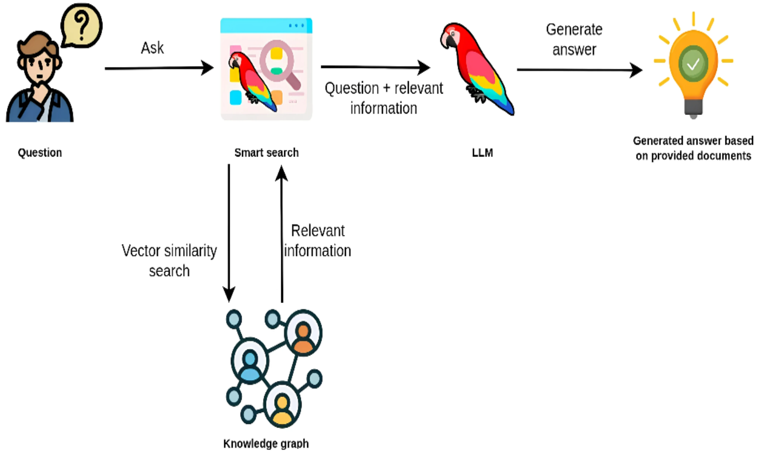
<Drug Discovery>



<Recommender System>



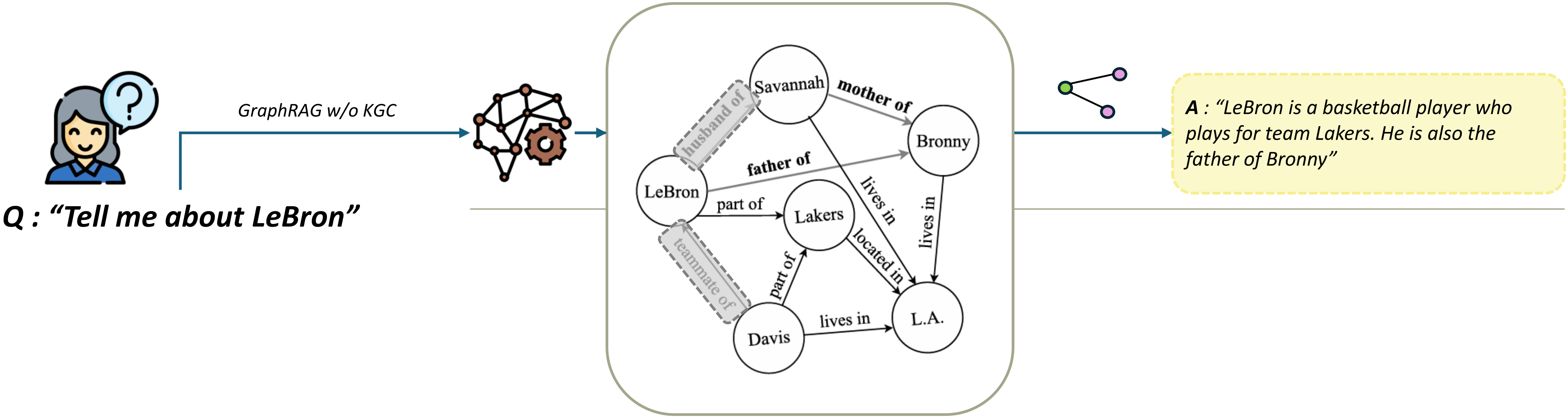
<GraphRAG>



KNOWLEDGE GRAPH COMPLETION

■ Missing Facts in KGs

- ☐ In Freebase and DBpedia, more than 66% of the person entities are missing a birthplace
- ☐ Greatly hinders the performance of systems that rely upon it

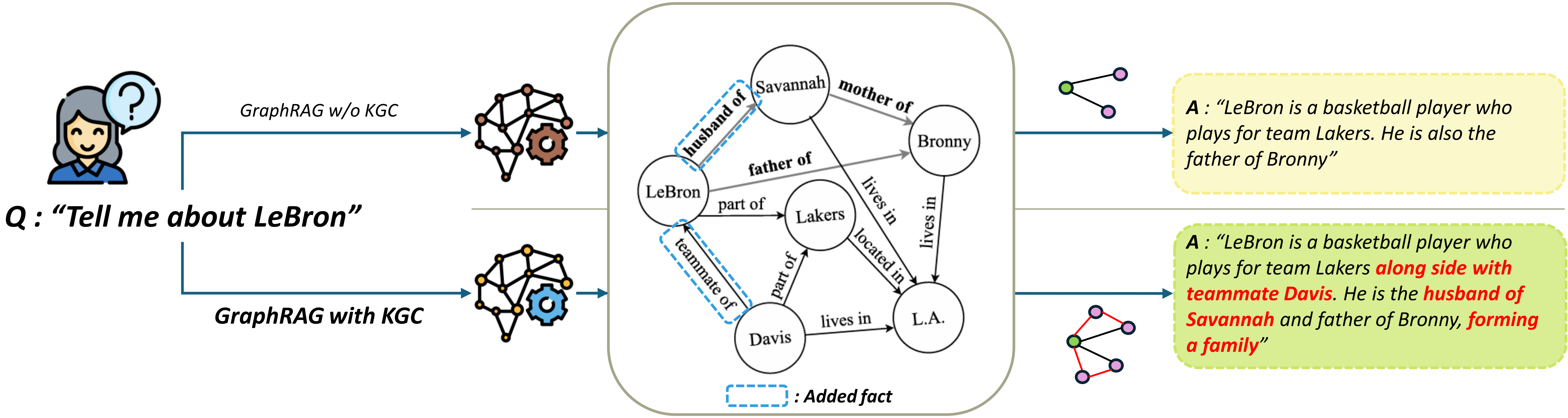


KNOWLEDGE GRAPH COMPLETION

■ Knowledge Graph Completion (KGC)

- ☐ KGC model to automatically fill missing links has been extensively studied
- ☐ Normally two tasks exist, link prediction(entity prediction) and relation prediction

link prediction : (h, r, ?) or (?, r, t) → predict "?"

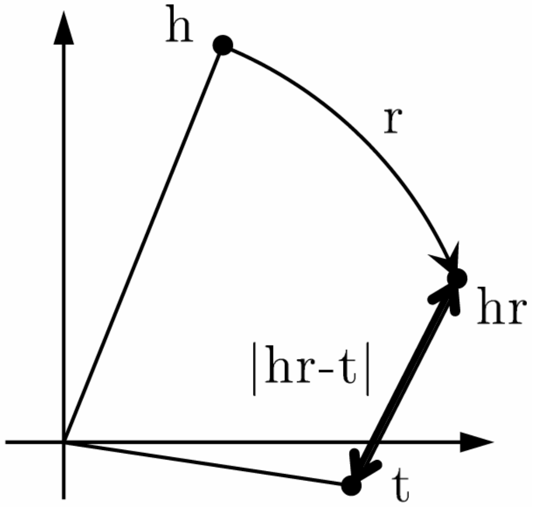


KNOWLEDGE GRAPH COMPLETION

■ KGC Expression & Categories

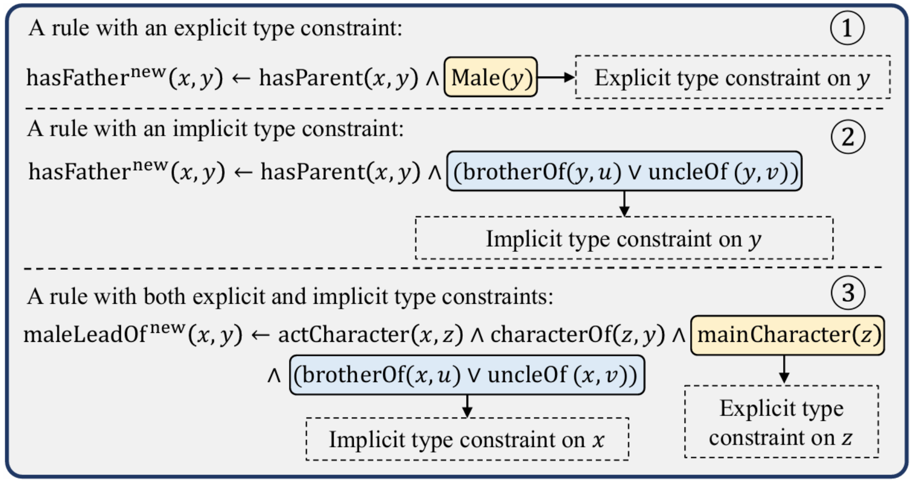
$(h, r, t) \in G^{tst}$ → *input* → S_θ → *output* → $score \in \mathbb{R}$

Embedding Based



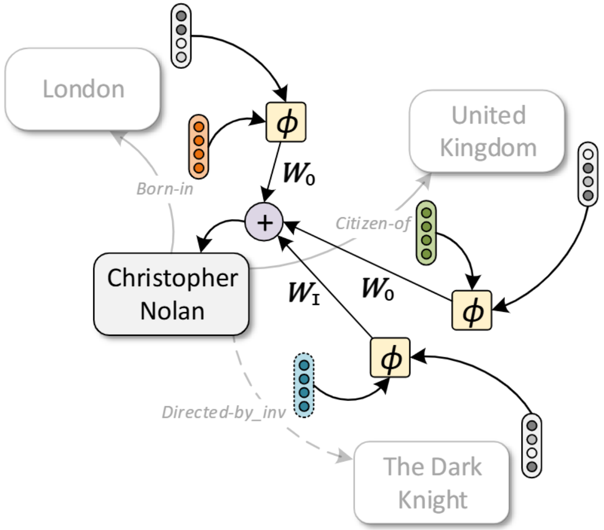
RotatE (ICLR'19)

Rule-Mining Based



TCLM (CIKM'24)

GNN Based



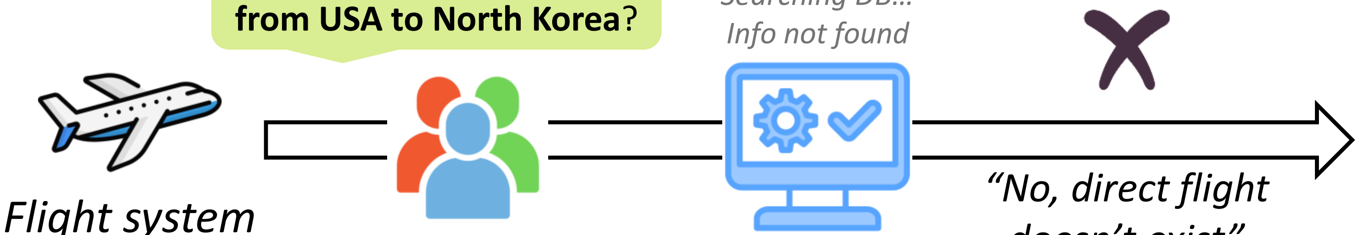
CompGCN (NeurIPS'20)

KGC EVALUATION & MOTIVATION

■ Inherent and Philosophical Question

Is there a **direct flight from USA to North Korea**?

Searching DB…
Info not found



Flight system

"No, direct flight doesn't exist"

Closed World Assumption

fact not present → **fact false**

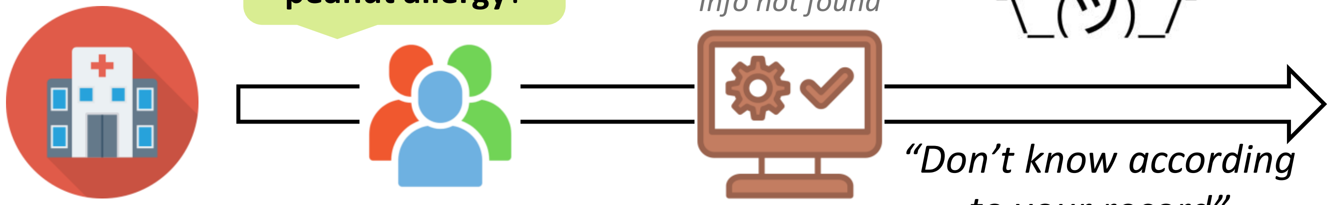
system knowledge is <u>complete</u> = the world is <u>**closed**</u>

Accuracy
F_1
ROC-AUC

Do I have **peanut allergy**?

Searching DB…
Info not found

¯_(ツ)_/¯



Medical system

"Don't know according to your record"

Open World Assumption

fact not present → **fact unknown**

System knowledge is <u>incomplete</u> = the world is <u>**open**</u>

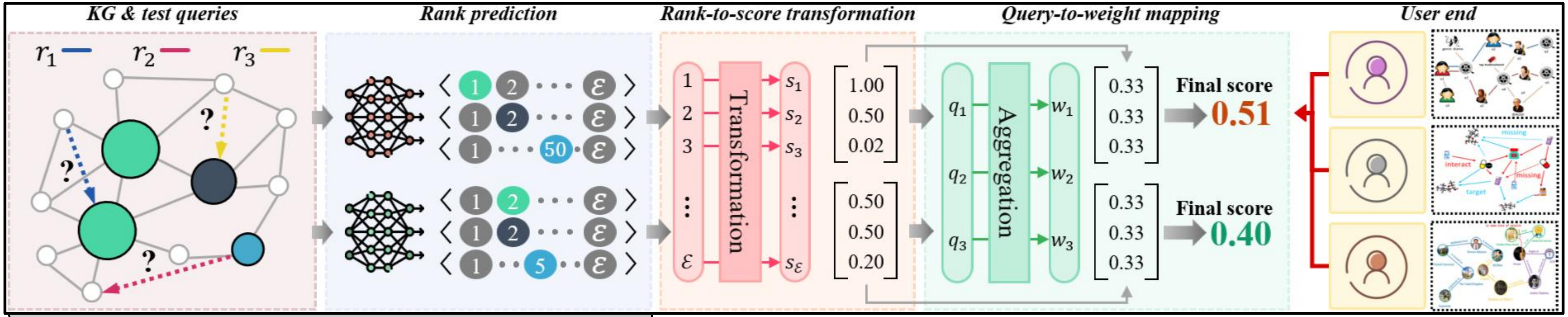
Rank based evaluation

KGC EVALUATION & MOTIVATION

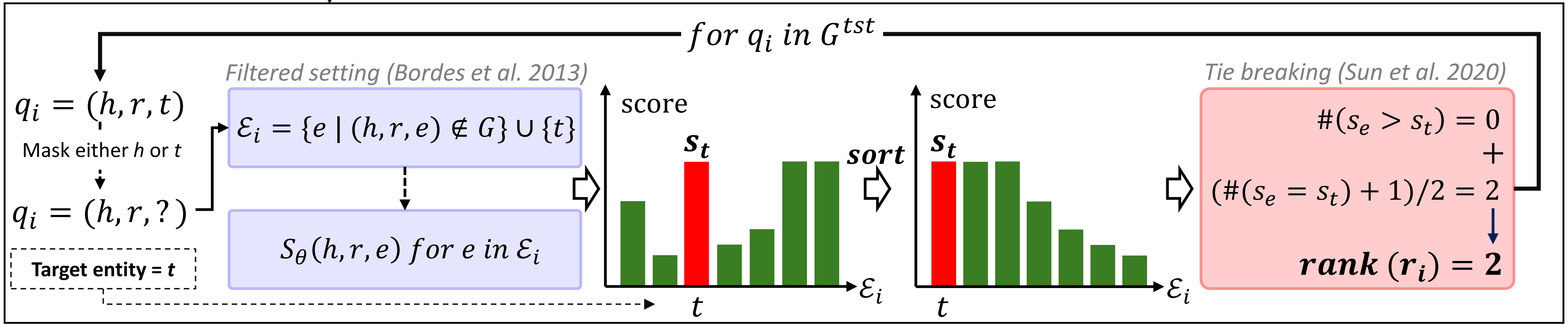
■ Overview of Rank Based Metrics

$(h, r, t) \in G^{tst}$ → *input* → S_{θ} → *output* → $score \in \mathbb{R}$

Metric Calc.



Rank Prediction



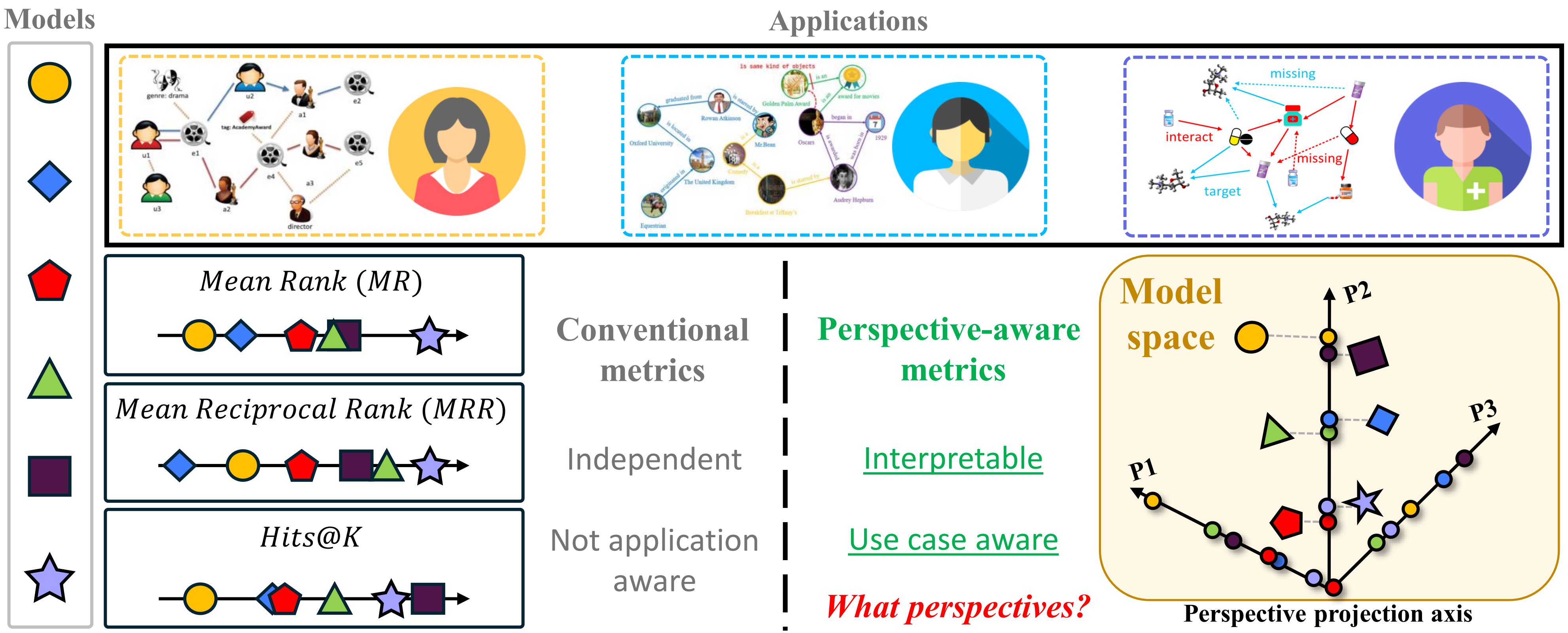
Bordes, Antoine, et al. "Translating embeddings for modeling multi-relational data." *Advances in neural information processing systems* 26 (2013).

Sun, Zhiqing, et al. "A re-evaluation of knowledge graph completion methods." *Proceedings of the 58th annual meeting of the association for computational linguistics*. 2020.

KGC EVALUATION & MOTIVATION

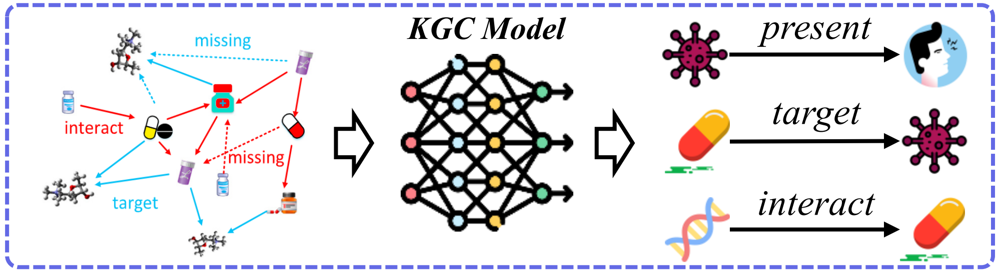
■ Shortcomings of Conventional Metrics

- □ Each practitioner has different characteristic, use of KG
- □ Multi-perspective evaluation is crucial for application-aware model selection



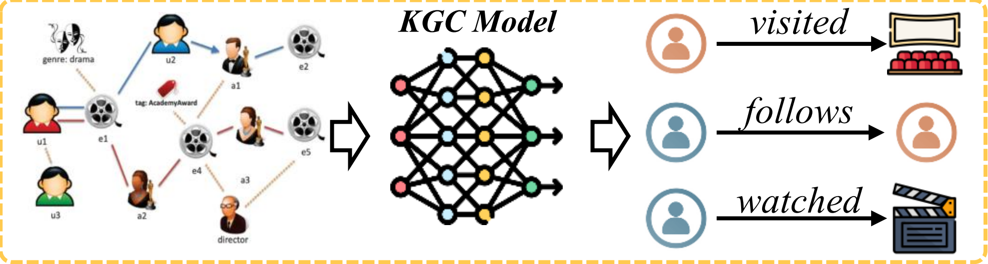
KGC EVALUATION & MOTIVATION

■ (P1) Predictive Sharpness



Drug prediction

"Clinical trials take years and millions of USD [1], <u>we can't tolerate less faithful facts</u>"
*Mostly interested in **top-rank prediction***

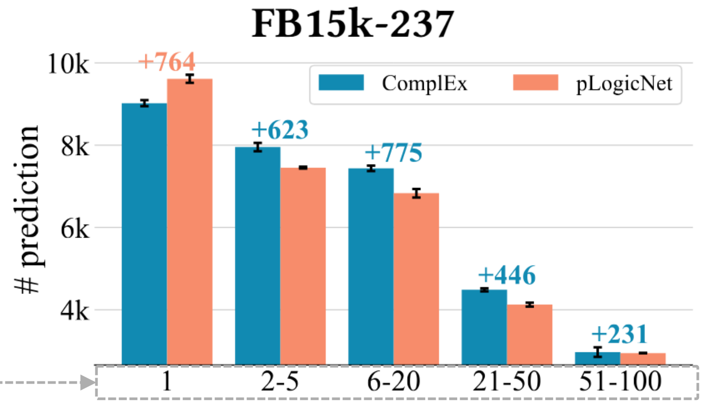


Rec. system

"As long as the KG captures meaningful relationship between user and item, <u>the fact is acceptable</u>"
*More generous to **less likely prediction***

Example case

Rank intervals



pLogicNet

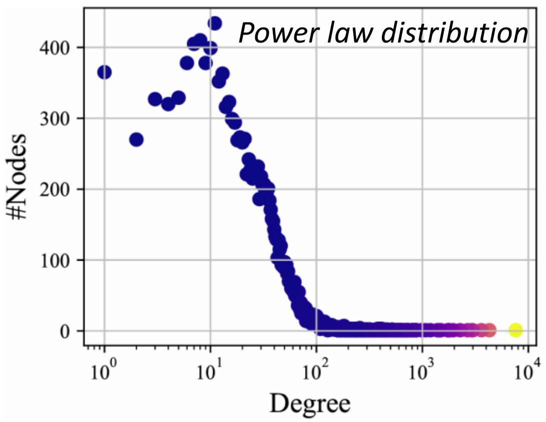
ComplEx

[1] Zeng, Xiangxiang, et al. "Toward better drug discovery with knowledge graph." *Current opinion in structural biology* 72 (2022): 114-126.

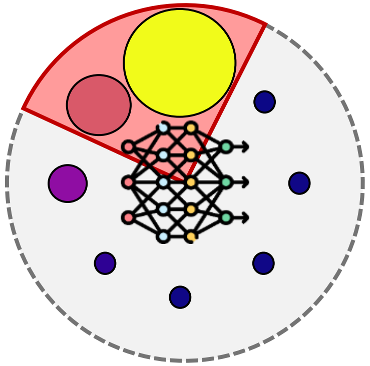
KGC EVALUATION & MOTIVATION

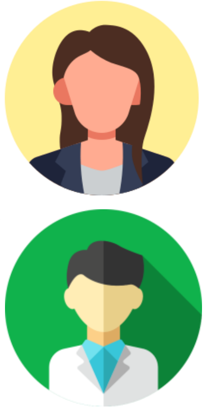
■ (P2) Popularity Bias Robustness

What bias?



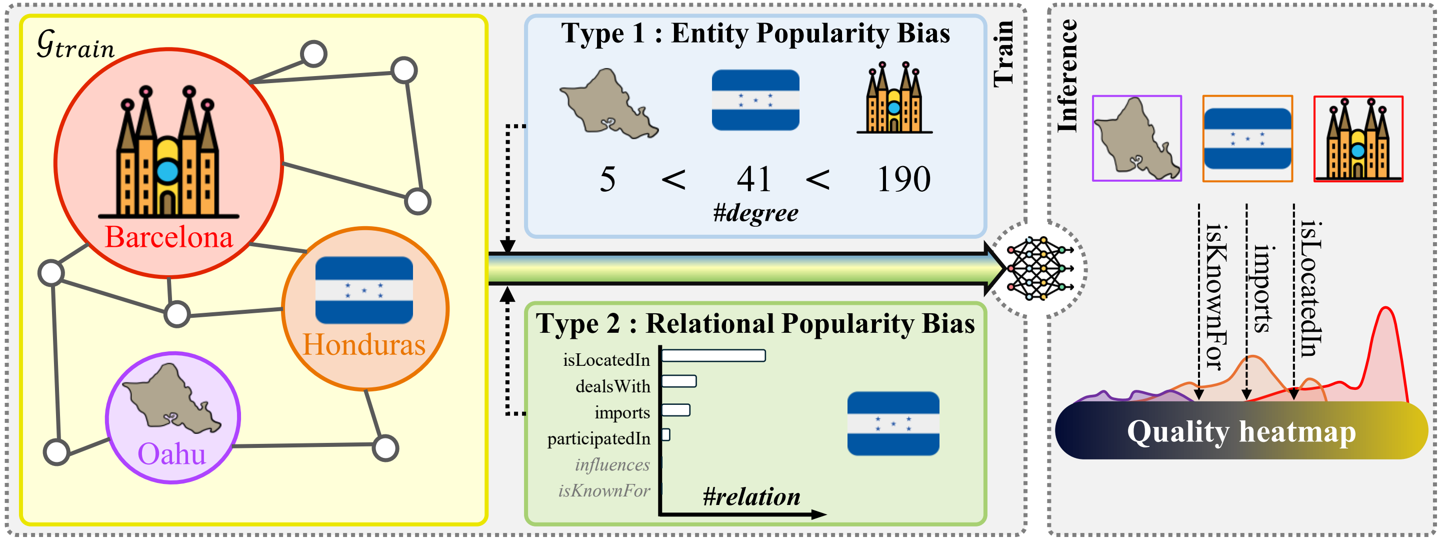
Pareto principle





"We treat all facts equally. "
*Interested in **pure precision***

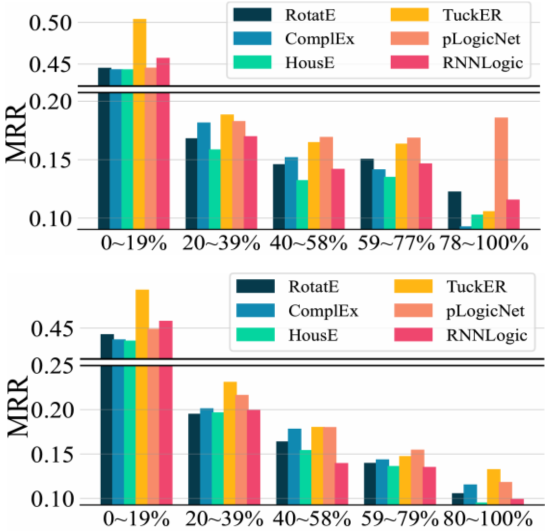
"Fresh and not so obvious facts convey more value"
*Interested in discovering **minor facts***



Type 1

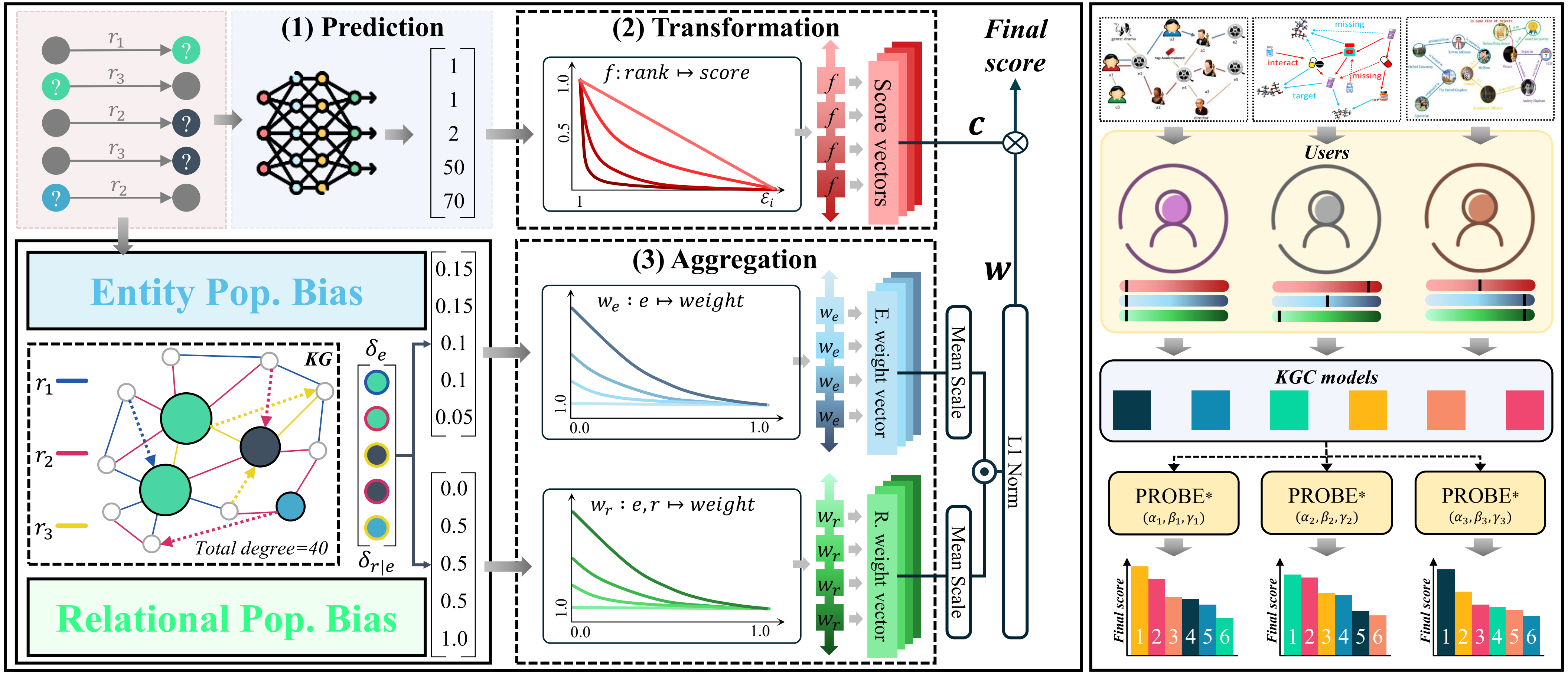
⇨ Reality

Type 2



KGC EVALUATION & MOTIVATION

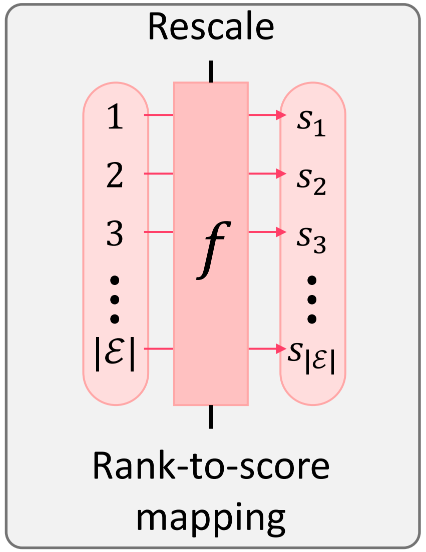
■ Overview of PROBE*



PREDICTIVE SHARPNESS

■ Rank Transformation (f) : Scoring Ranks

Rank Transformation



Sharpness control factor (α)

$$f(r,\alpha) = r^{-\alpha}$$

$\alpha = -1$ → MR

$\alpha = 1$ → MRR

Evaluation

Tiwari et al., Hoyt et al.

Unstable

↑

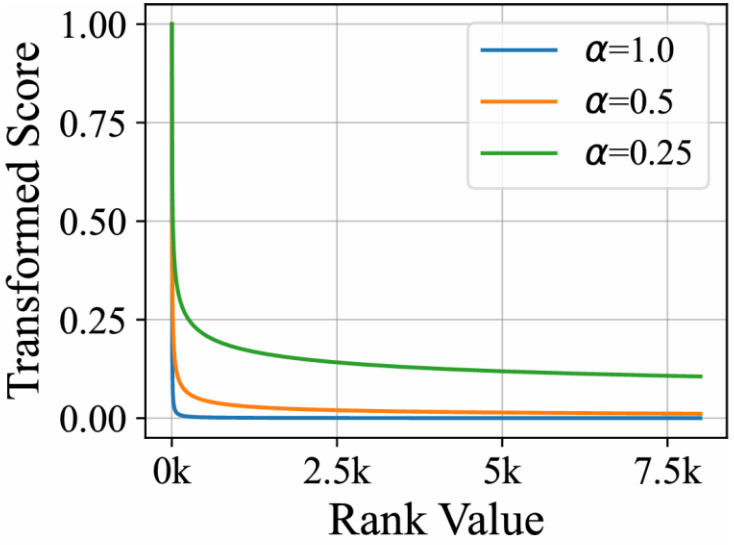
$\alpha < 0 \rightarrow f \in [1, |\mathcal{E}|^{-\alpha}]$

$\alpha > 0 \rightarrow f \in [|\mathcal{E}|^{-\alpha}, 1]$

↓

Stable

f

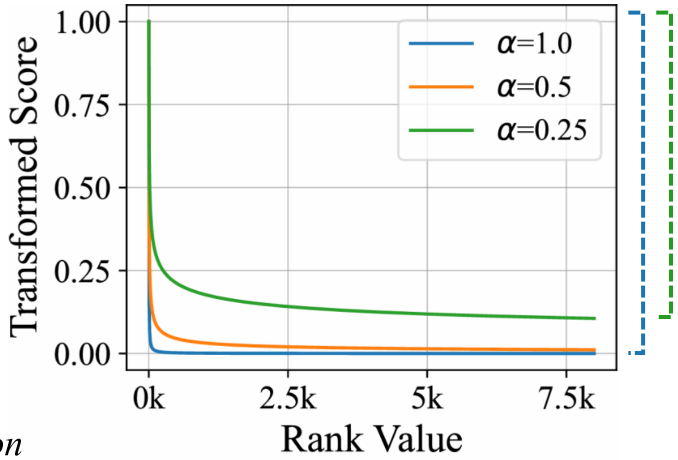


[1] Tiwari, Sudhanshu, Iti Bansal, and Carlos R. Rivero. "Revisiting the evaluation protocol of knowledge graph completion methods for link prediction." *Proceedings of the Web Conference 2021*.
[2] Hoyt, Charles Tapley, et al. "A unified framework for rank-based evaluation metrics for link prediction in knowledge graphs." *arXiv preprint arXiv:2203.07544* (2022).

PREDICTIVE SHARPNESS

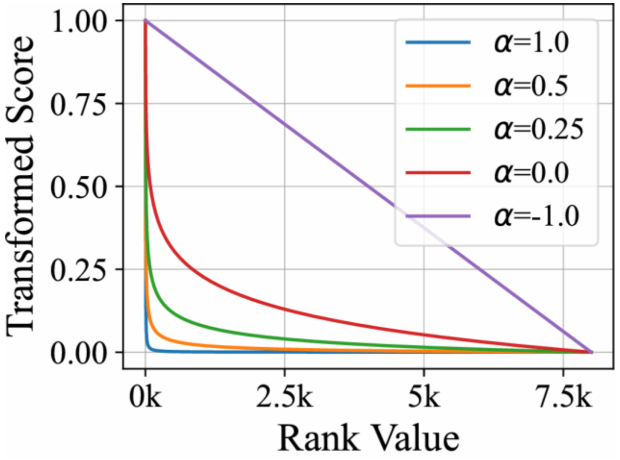
■ Subtle, Yet Crucial Details

$$f(r, \alpha) = r^{-\alpha}$$



Affine transformation

$$f*(r_i, \alpha) = \frac{r_i^{-\alpha} - \mathcal{E}_i^{-\alpha}}{1 - \mathcal{E}_i^{-\alpha}}$$



Problem

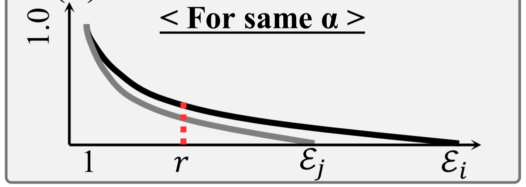
Compressed upper and lower bound

⇩

Solution

Securing absolute lower bound with $\mathcal{E}_i$

(1) Candidate size aware



(2) Expanding range of α

$[0, \infty] \Rightarrow [-1, \infty]$

$$\lim_{\alpha \to 0} f^*(r_i, \alpha) = 1 - \frac{\ln r_i}{\ln \mathcal{E}_i}$$

$[2, 1, 10, \cdots, 19, 31, 6]$

$f*$

Score vector

Candidate sample of i − th query : $\mathcal{E}_i$

PREDICTIVE SHARPNESS

(A1) Interpreting Related Works Under PROBE*

Lemma 1. $Hits@1\ is\ equal\ to\ PROBE *_{\alpha \to \infty}$

Lemma 2. $PROBE *_{\alpha = -1}\ is\ a\ chance\ adjusted\ metric$

<Adjusted metric line of work>

$\mathfrak{M} \to \overline{\mathfrak{M}} \in [-1,1]$
- 1 (optimum)
- 0 (random, $\mathbb{E}[\overline{\mathfrak{M}}] = 0$)
- -1 (pessimum)

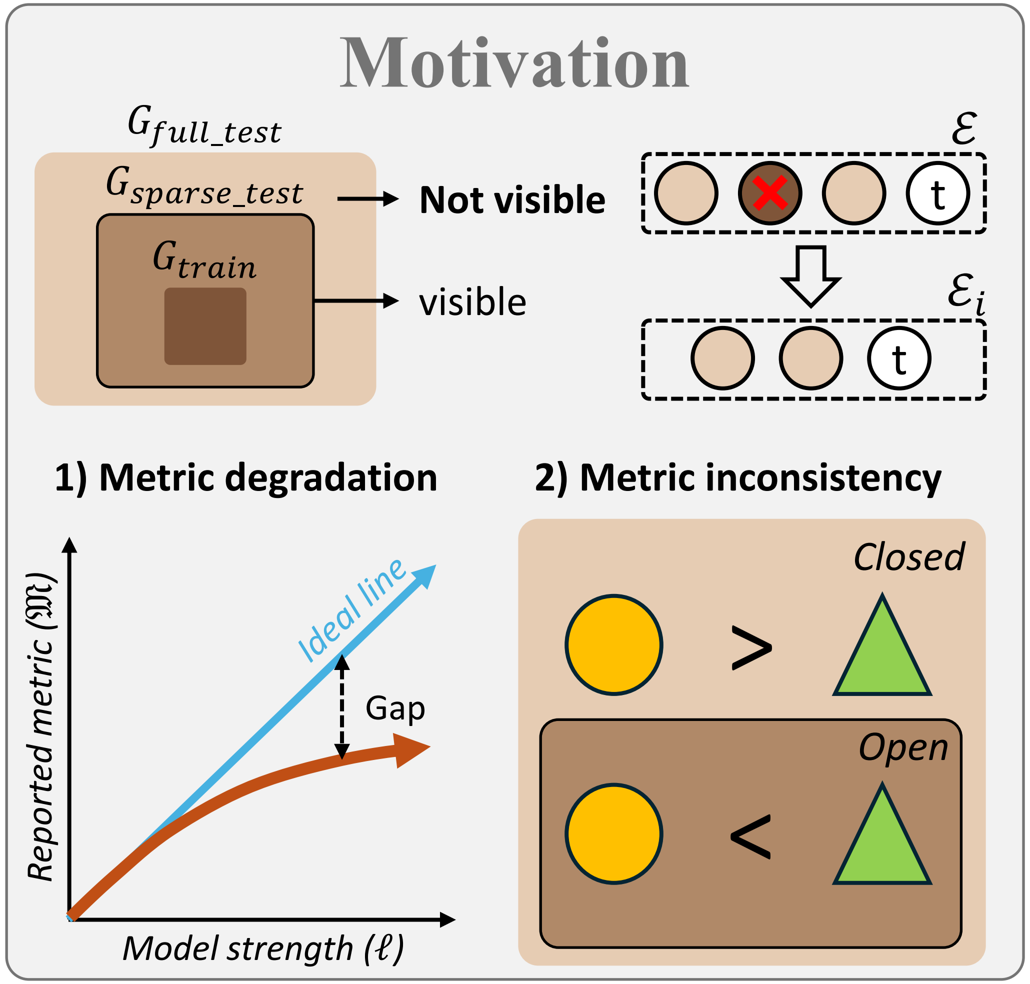
<PROBE* from the adjusted view>

- 1 (optimum)
- 0.5 (random, $\mathbb{E}[\text{PROBE} *] = 0.5$)
- 0 (pessimum)

Proofs in appendix A

PREDICTIVE SHARPNESS

■ (A2) Interpreting Under the Open World (Extending Yang et al.)



Yang's Analysis

$$g(r) = rf(r)$$

$$\frac{d\hat{\mathbb{E}}(\mathfrak{M})}{d\ell} = \frac{1}{l\beta(N+1)} \mathbb{E}_{k\sim\mathcal{B}(N+1,\ell\beta)}\, g(k)$$

↓ derivative → ↓ degradation & ↓ inconsistency

More credibility to the conclusion

Lemma 3. α *strictly decreases PROBE* $*$ expectation w. r. t model strength

Yang, Haotong, Zhouchen Lin, and Muhan Zhang. "Rethinking knowledge graph evaluation under the open-world assumption." *Advances in Neural Information Processing Systems* 35 (2022): 8374-8385.

Proof in appendix B

PREDICTIVE SHARPNESS

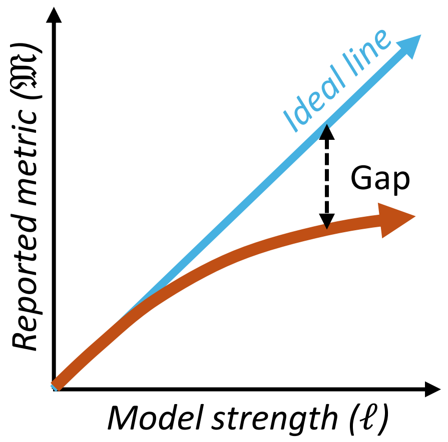
(A2) Experiments Under CW and OW

Family-tree Dataset

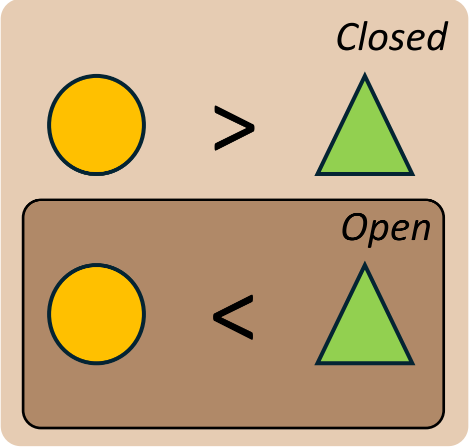
# entities	6,004
# relations	23
# facts	192,532

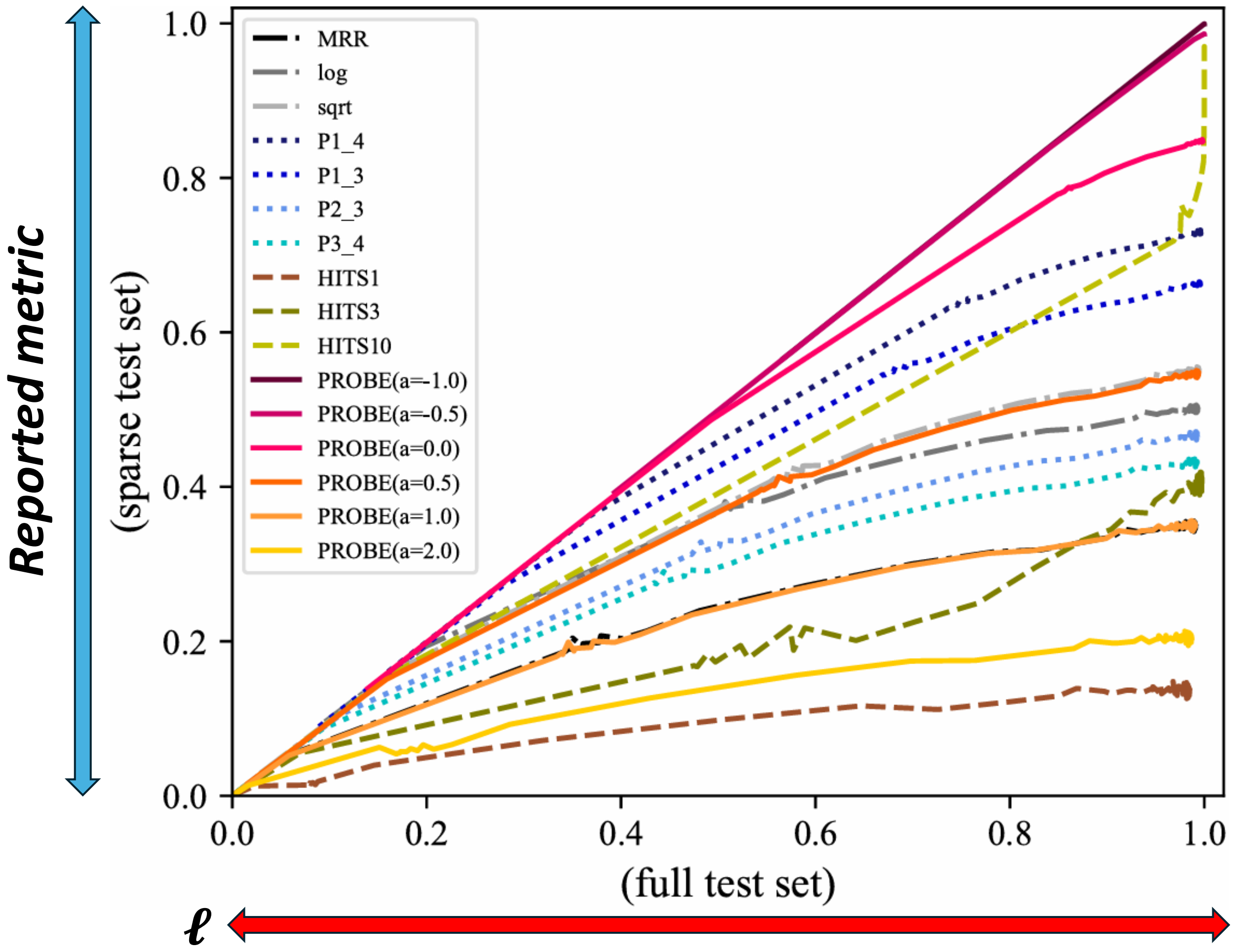
- Closed-world
- Sparsity controllable (sparsity is set to 25%)

1) Metric degradation



2) Metric inconsistency





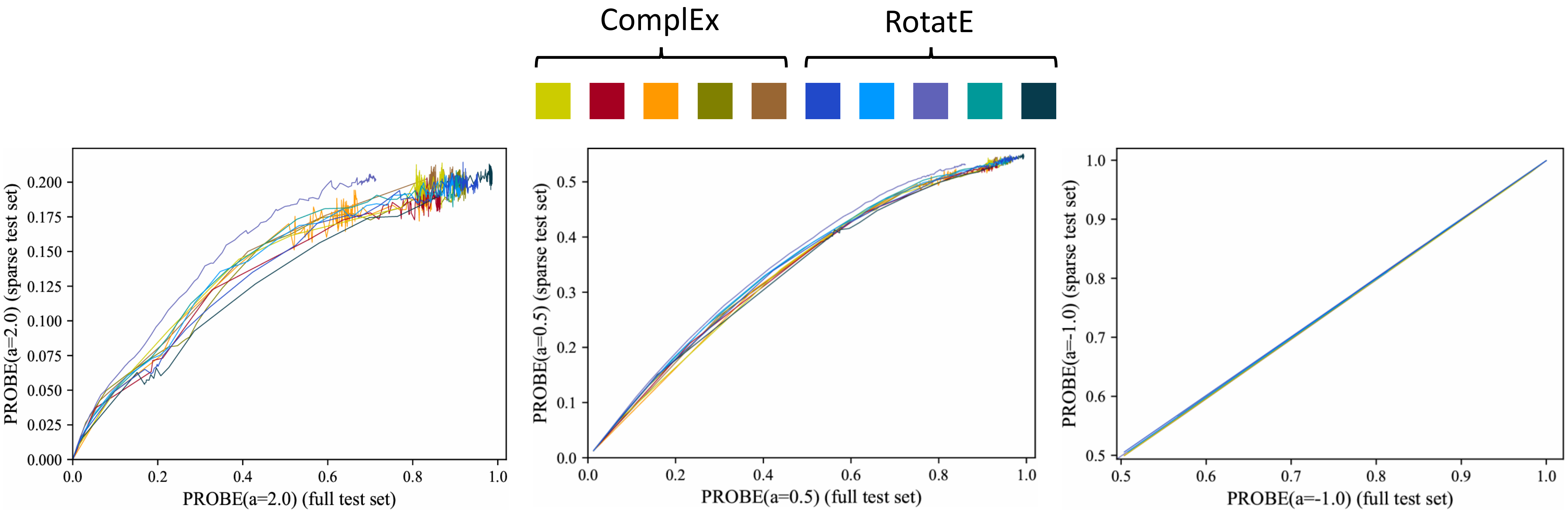
Progress bar

PREDICTIVE SHARPNESS

■ (A2) How Should We View It

Low sharpness is the best?! Counter example : $\mathfrak{M} = 1$

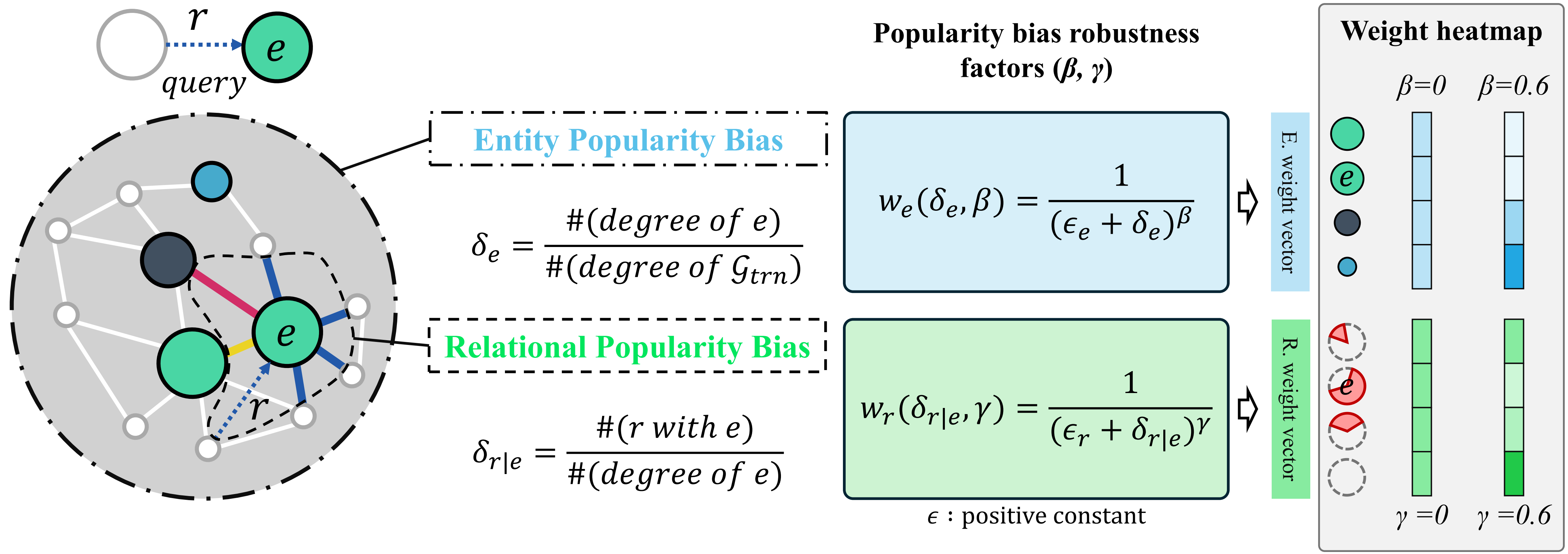
What we can take How PROBE* reacts to the open world



Other plots in appendix C

POPULARITY BIAS ROBUSTNESS

■ Global Entity Bias & Local Relational Bias



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POPULARITY BIAS ROBUSTNESS

■ Combining Two Weights

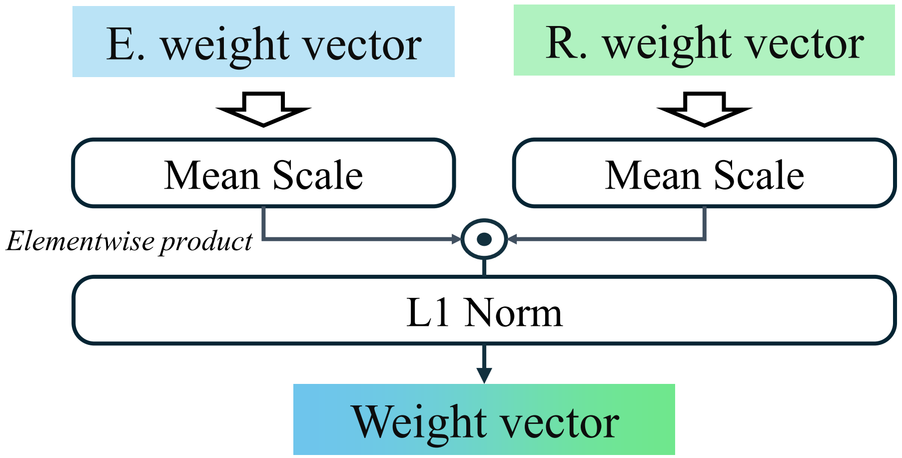
Problem

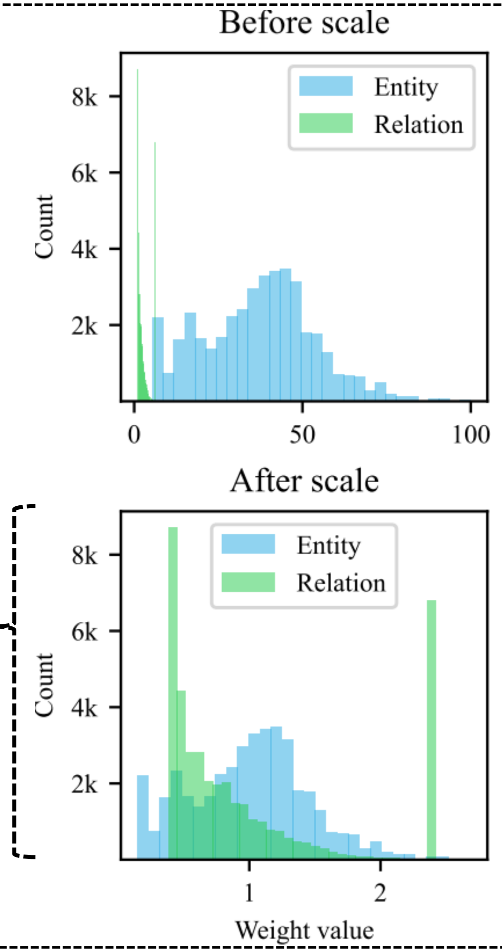
Two vectors have completely different scale.
One weight can dominate the other.

⇩

Solution

Mean scale two vectors





Summary of PROBE*

c Score vector

•

w Weight vector

↓

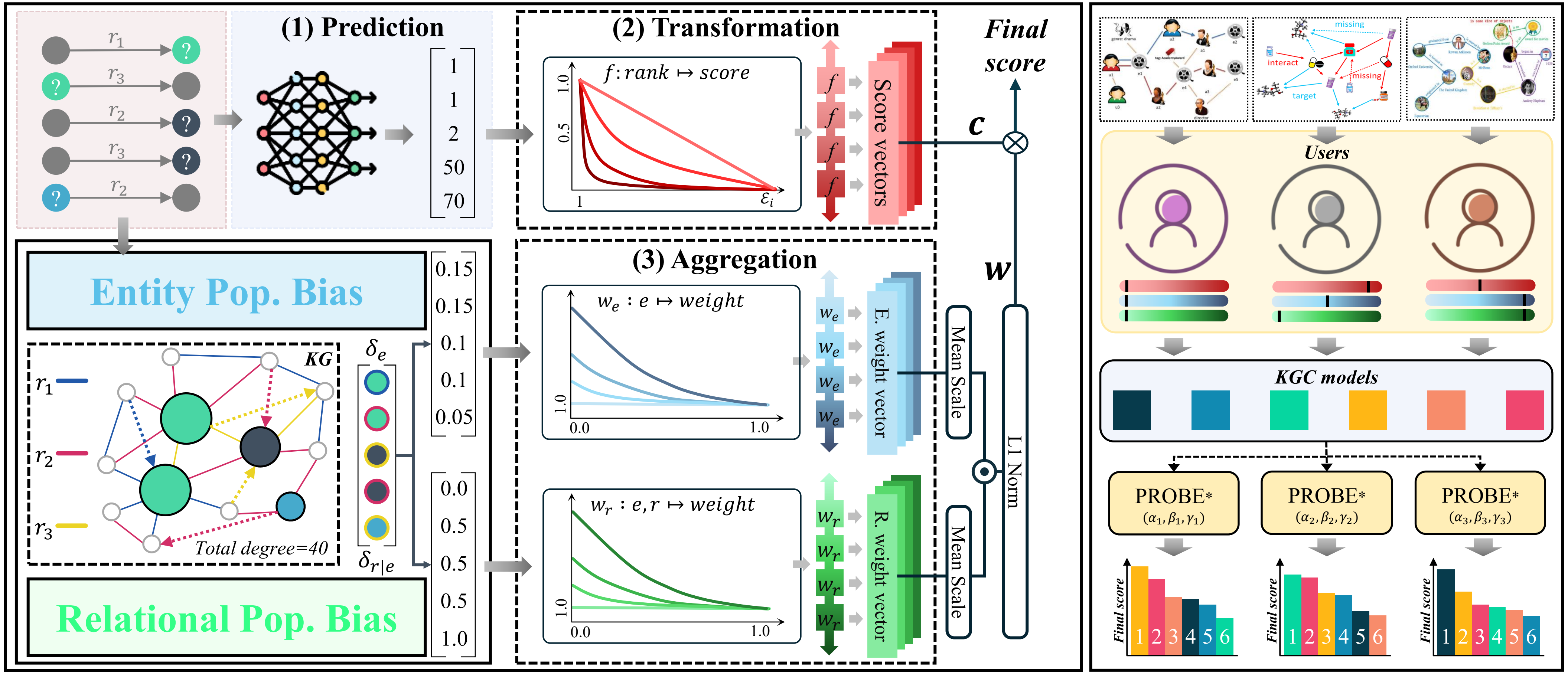
Final score

PROBE* is the **dot product** between score and weight vector

COMBINING PERPECTIVES

■ PROBE* Architecture in a Netshell

Background & Motivation → Framework & Analysis → Experiments



Progress bar

COMBINING PERPECTIVES

■ Desired Properties of Rank Based Metric for KGC

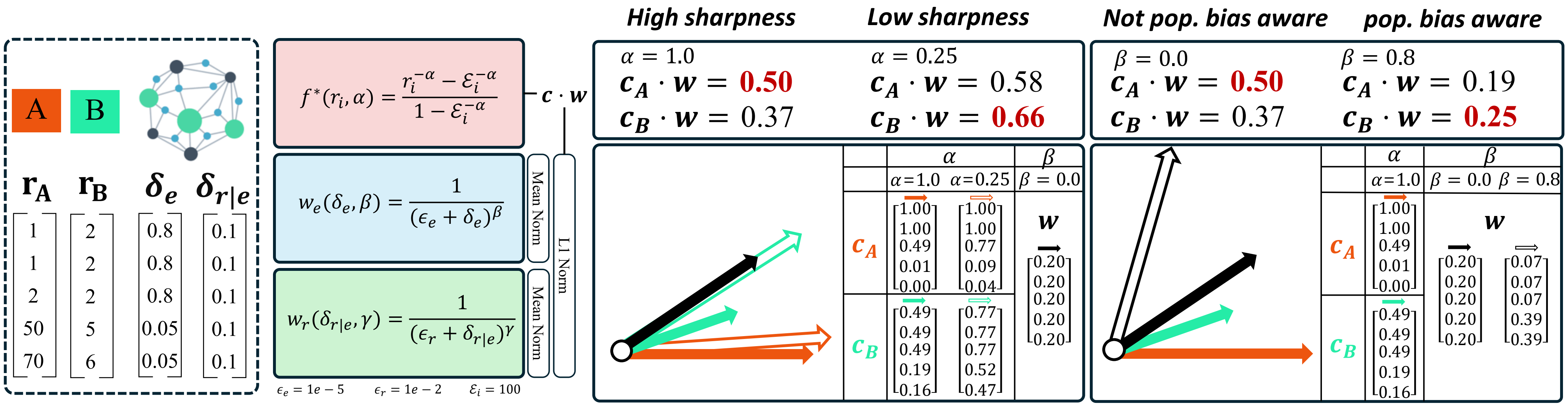
Property	Constraint	MR	MRR	Hits@K	p-MRR	Strat-MRR	WMR	AMRI	**PROBE***
Fixed optimum	$\mathfrak{M}(\mathbf{1}_{\lvert Q\rvert}) = c_{opt}$	✓	✓	✓	✓	✓	✓	✓	✓
Fixed pessimum	$\mathfrak{M}(\mathcal{E}_{\lvert Q\rvert}) = c_{pes}$	-	-	✓	-	-	-	✓	✓
Anti-monotonic	$r_i > r_j \longrightarrow \mathfrak{M}(r_i) < \mathfrak{M}(r_j)$	-	✓	-	✓	✓	-	✓	✓
Counterpart-aware	$\mathcal{E}_1 \neq \mathcal{E}_2 \longrightarrow \mathfrak{M}_1(\mathbf{r}) \neq \mathfrak{M}_2(\mathbf{r})$	-	-	-	-	-	✓	✓	✓
Sharpness-aware	$\alpha \neq \alpha' \longrightarrow \mathfrak{M}_\alpha(\mathbf{r}) \neq \mathfrak{M}_{\alpha'}(\mathbf{r})$	-	-	-	✓	-	-	-	✓
Popularity-aware	$w_q \propto \delta(q)$	-	-	-	-	-	-	-	✓

$\mathfrak{M}$	MR	MRR	Hits@K	p-MRR	Strat-MRR	WMR	AMRI	PROBE*
Def	$\frac{1}{\lvert Q\rvert}\sum_i r_i$	$\frac{1}{\lvert Q\rvert}\sum_i \frac{1}{r_i}$	$\frac{1}{\lvert Q\rvert}\sum_i \mathbb{I}[r_i \leq K]$	$\frac{1}{\lvert Q\rvert}\sum_i r_i^{-p}$	$\frac{\sum_{rel} w^{rel} strat-MRR_{rel}}{\sum_{rel\in\mathcal{R}} w^{rel}}$	$\exp\left(\frac{\sum_i(\mathcal{E}_i-1)\ln r_i}{\sum_i(\mathcal{E}_i-1)}\right)$	$1-\frac{2\sum_i(r_i-1)}{\sum_i(\mathcal{E}_i-1)}$	$\mathbf{c}\cdot\mathbf{w}$

COMBINING PERPECTIVES

■ How Final Scores Change with Perspective Shift

PROBE* is the **dot product** between score and weight vector



EXPERIMENTS

Models, Datasets & Implementation Details

- RotatE
- ComplEx
- HousE
- TuckER
- pLogicNet
- RNNLogic

- Low dimension (d=128) tuned
- Filtered setting, tie breaking protocol adopted for fair evaluation
- Results are average of 3

α = [1.0, 0.5, 0.0, -0.5, -1.0]
β = [0.0, 0.2, 0.4, 0.6, 0.8]
γ = [0.0, 0.2, 0.4, 0.6, 0.8]

$\epsilon_e = 1e - 5$

$\epsilon_r = 1e - 2$

Dataset	$\lvert\mathcal{E}\rvert$	$\lvert\mathcal{R}\rvert$	$\lvert\mathcal{G}_{trn}\rvert$	$\lvert\mathcal{G}_{val}\rvert$	$\lvert\mathcal{G}_{tst}\rvert$
FB15k237	14,541	237	272,115	17,535	20,466
WN18RR	40,943	11	86,835	3,034	3,134
YAGO3-10	123,182	37	1,079,040	5,000	5,000
Family	3,007	12	23,483	2,038	2,835
UMLS	135	46	1,959	1,306	3,264
Kinship	104	25	3,206	2,137	5,343

EXPERIMENTS

■ Perspective Wise Tables

Models	FB15k-237					WN18RR					YAGO3-10				
	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0
RotatE	0.3137	0.4239	0.7124	0.938	0.9844	0.4649	0.5217	0.6945	0.8537	0.9006	0.4729	0.5625	0.7966	0.9554	0.9812
ComplEx	0.3135	0.4274	0.7209	0.9424	0.9854	0.4752	0.5183	0.661	0.815	0.873	0.4447	0.5426	0.7957	**0.9577**	0.9807
HousE	0.3073	0.41	0.691	0.9244	0.9778	0.4184	0.4913	0.6991	0.8823	**0.9346**	0.4538	0.5297	0.75	0.9267	0.9647
TuckER	**0.353**	**0.4608**	**0.7379**	**0.9479**	**0.9884**	0.4625	0.4999	0.6368	0.7961	0.8583	**0.5654**	**0.6319**	**0.8176**	0.9561	0.9814
pLogicNet	0.3241	0.4311	0.7149	0.9386	0.9847	**0.4957**	**0.5672**	**0.7593**	**0.9002**	0.9292	0.4535	0.5467	0.7927	0.9576	**0.9834**
RNNLogic	0.3197	0.4218	0.6969	0.9213	0.9723	0.4807	0.5225	0.6567	0.7929	0.8401	-	-	-	-	-

Models	Family					UMLS					Kinship				
	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0	α=1.0	α=0.5	α=0.0	α=-0.5	α=-1.0
RotatE	0.9478	0.9639	0.9832	0.9921	0.9938	0.7782	0.8369	0.9056	0.9556	0.9776	0.6318	0.7224	0.8365	0.9273	0.9717
ComplEx	0.9431	0.9616	0.9835	**0.9941**	**0.9966**	0.5993	0.6796	0.79	0.885	0.9362	0.5416	0.648	0.787	0.9015	0.9596
HousE	0.9757	0.9818	0.9893	0.9938	0.9953	0.8354	0.8775	0.9271	0.9637	0.9802	0.6808	0.762	0.8621	0.94	**0.977**
TuckER	**0.9784**	**0.9832**	**0.9895**	0.9939	0.9959	**0.8439**	**0.8849**	**0.9323**	**0.9669**	**0.9825**	**0.7067**	**0.7808**	**0.8716**	**0.9421**	0.9759
pLogicNet	0.8119	0.86	0.9355	0.9837	0.9939	0.5781	0.6608	0.7748	0.8764	0.9346	0.3771	0.4501	0.5628	0.6803	0.7641
RNNLogic	0.9737	0.9781	0.9839	0.9882	0.9901	0.7942	0.8455	0.9068	0.9526	0.9739	0.6417	0.7289	0.8394	0.9277	0.9712

EXPERIMENTS

■ Case Study : Predictive Sharpness

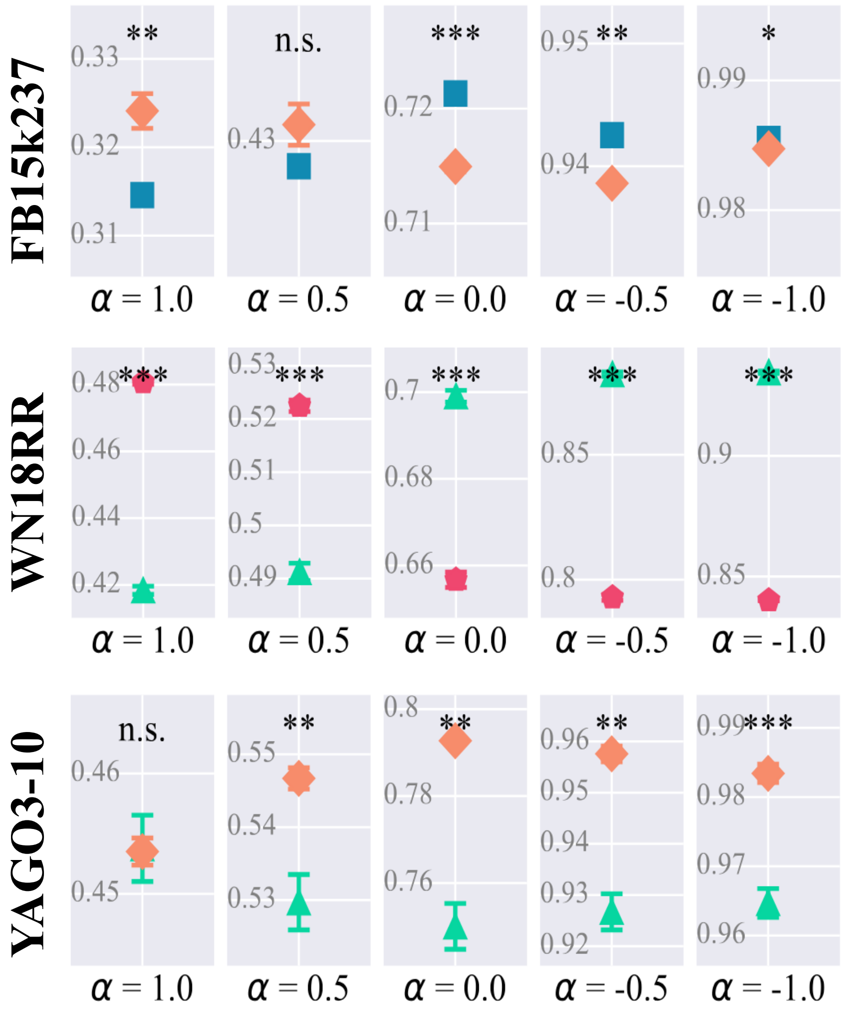
Results with statistical significance

p ≥ 5e-2 (n.s.)

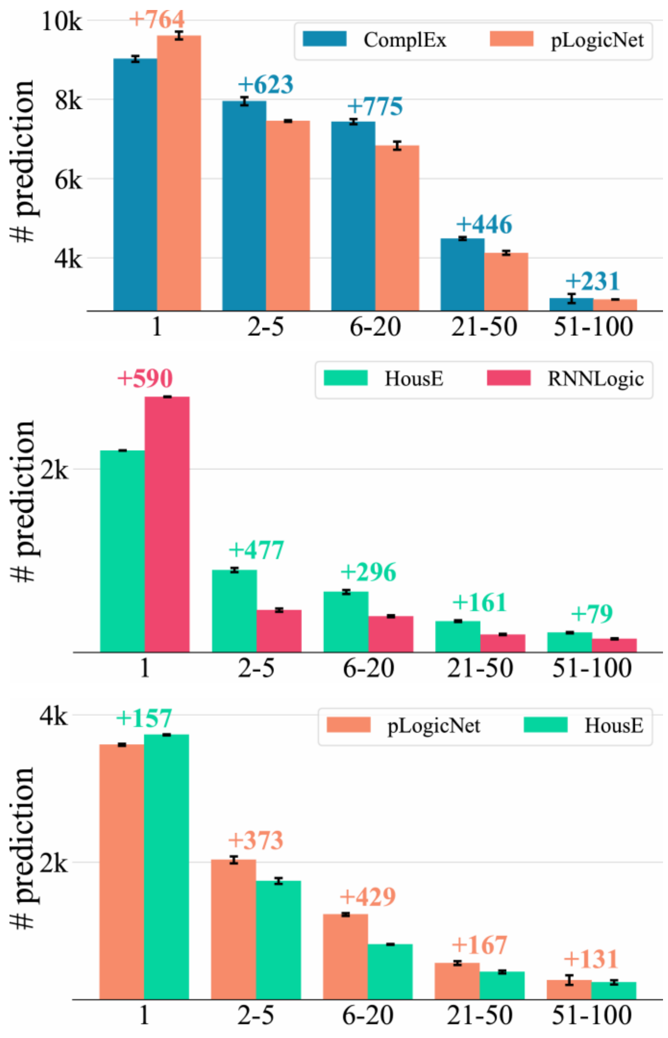
p < 5e-2 (*)

p < 1e-2 (**)

p < 1e-3 (***)



Why?



EXPERIMENTS

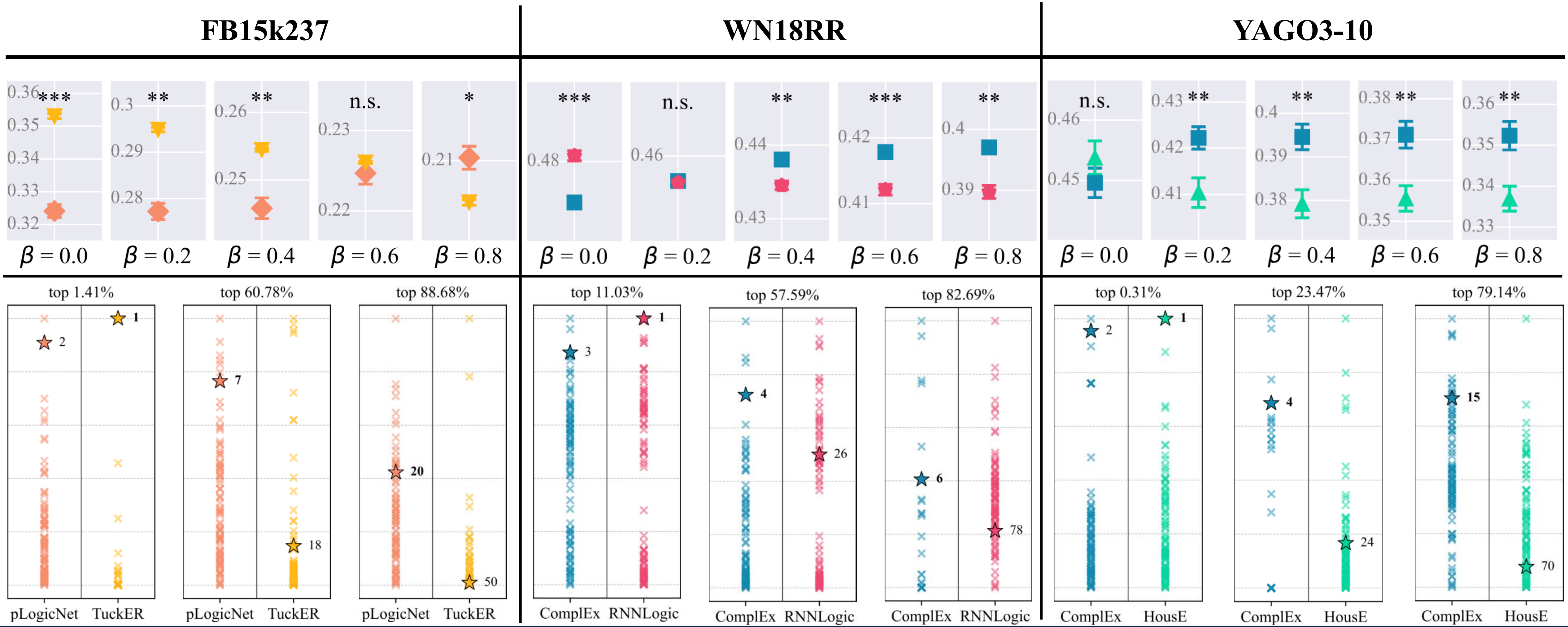
Perspective Wise Tables

Models	FB15k-237					WN18RR					YAGO3-10				
	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8
RotatE	0.314	0.2633	0.2286	0.2045	0.1873	0.4641	0.4485	0.4318	0.4141	0.3957	0.4743	0.4288	0.3976	0.3747	0.3572
ComplEx	0.3146	0.2662	0.2319	0.2072	0.1886	0.4754	0.4574	0.4381	**0.4178**	**0.3971**	0.4459	0.418	0.39	0.3665	0.3475
HousE	0.3073	0.2533	0.2167	0.1916	0.1737	0.4184	0.4038	0.3897	0.3754	0.3607	0.4538	0.4104	0.3793	0.3556	0.337
TuckER	**0.353**	**0.2948**	**0.2544**	**0.226**	0.2053	0.4625	0.4426	0.4225	0.4022	0.3816	**0.5654**	**0.5178**	**0.4842**	**0.4589**	**0.4391**
pLogicNet	0.3241	0.2772	0.2458	0.2247	**0.2103**	**0.4957**	**0.4657**	**0.4388**	0.4143	0.3916	0.4535	0.4091	0.3791	0.3575	0.3414
RNNLogic	0.3197	0.2641	0.2266	0.2011	0.1831	0.4807	0.4572	0.4345	0.4121	0.3897	-	-	-	-	-

Models	Family					UMLS					Kinship				
	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8	β=0.0	β=0.2	β=0.4	β=0.6	β=0.8
RotatE	0.9487	0.9404	0.9278	0.9084	0.879	0.7784	0.7625	0.7363	0.6845	0.5736	0.6336	0.633	0.6325	0.6319	0.6313
ComplEx	0.9434	0.939	0.9307	0.9158	0.8908	0.6046	0.6047	0.601	0.5875	0.5522	0.5394	0.5389	0.5383	0.5377	0.5372
HousE	0.9757	0.9689	0.9572	0.938	0.9078	0.8354	0.8229	0.7996	0.7482	0.6315	0.6808	0.6802	0.6795	0.6789	0.6783
TuckER	**0.9784**	**0.9719**	**0.9608**	**0.9426**	**0.9138**	**0.8439**	**0.8318**	**0.8129**	**0.779**	**0.7143**	**0.7067**	**0.7058**	**0.7049**	**0.7041**	**0.7032**
pLogicNet	0.8119	0.8026	0.7896	0.7714	0.7462	0.5781	0.5786	0.5741	0.5553	0.5009	0.3771	0.3768	0.3765	0.3762	0.3759
RNNLogic	0.9737	0.9647	0.9506	0.9286	0.8957	0.7942	0.7793	0.7541	0.7037	0.5967	0.6417	0.6408	0.6398	0.6388	0.6379

EXPERIMENTS

■ Case Study (Entity Bias) : Top-100 Prediction Scores



EXPERIMENTS

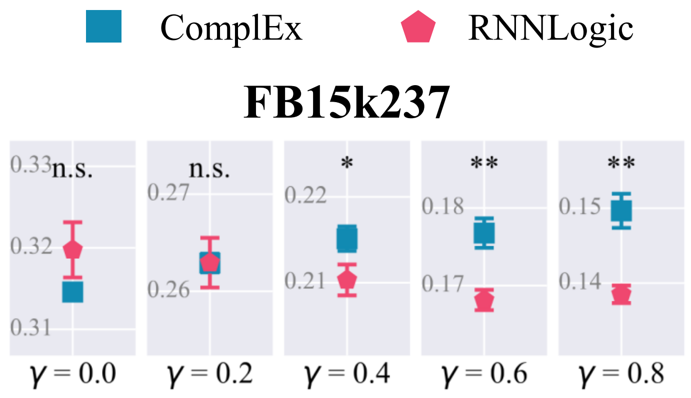
Perspective Wise Tables

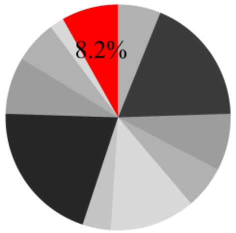
Models	FB15k-237					WN18RR					YAGO3-10				
	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8
RotatE	0.314	0.2603	0.2104	0.1702	0.1417	0.4641	0.3854	0.2972	0.2244	0.1777	0.4743	0.4347	0.3698	0.2856	0.2031
ComplEx	0.3146	0.2629	0.2151	0.1768	0.1496	0.4754	0.3913	0.2969	0.2188	0.1685	0.4496	0.4156	0.3569	0.279	0.2016
HousE	0.3073	0.2535	0.2032	0.1623	0.1331	0.4184	0.3483	0.2704	0.2065	0.1657	0.4538	0.4115	0.3412	0.2497	0.1599
TuckER	**0.353**	**0.2961**	**0.2433**	**0.201**	**0.1713**	0.4625	0.3735	0.2738	0.1915	0.1386	**0.5654**	**0.5187**	**0.4414**	**0.3409**	**0.2422**
pLogicNet	0.3241	0.2726	0.224	0.1843	0.1558	**0.4957**	**0.4118**	**0.3198**	**0.245**	**0.1975**	0.4535	0.417	0.3565	0.2776	0.1997
RNNLogic	0.3197	0.2629	0.2103	0.1681	0.1385	0.4807	0.3897	0.2875	0.2031	0.1487	-	-	-	-	-

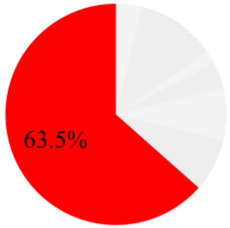
Models	Family					UMLS					Kinship				
	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8	γ=0.0	γ=0.2	γ=0.4	γ=0.6	γ=0.8
RotatE	0.9487	0.9484	0.9474	0.9458	0.9439	0.7784	0.771	0.7585	0.7418	0.7227	0.6336	0.6271	0.6193	0.6101	0.5996
ComplEx	0.9434	0.9462	0.9485	0.9503	0.9514	0.6046	0.6131	0.6179	0.6193	0.6178	0.5394	0.5354	0.5304	0.5242	0.5169
HousE	0.9757	0.9738	0.971	0.9678	0.9647	0.8354	**0.8269**	**0.8132**	**0.7952**	**0.7747**	0.6808	0.6705	0.6584	0.6444	**0.6287**
TuckER	**0.9784**	**0.9761**	**0.9731**	**0.9696**	**0.9663**	**0.8439**	0.8262	0.803	0.7755	0.7458	**0.7067**	**0.6905**	**0.6714**	**0.6495**	0.6251
pLogicNet	0.8119	0.7989	0.7845	0.7704	0.7583	0.5781	0.5615	0.5405	0.5165	0.4917	0.3771	0.3798	0.3822	0.3841	0.3854
RNNLogic	0.9737	0.9699	0.9649	0.9592	0.9537	0.7942	0.7776	0.7554	0.7288	0.7001	0.6417	0.6222	0.5993	0.5732	0.5444

EXPERIMENTS

Case Study (Relational Bias) : Relation Pie Chart

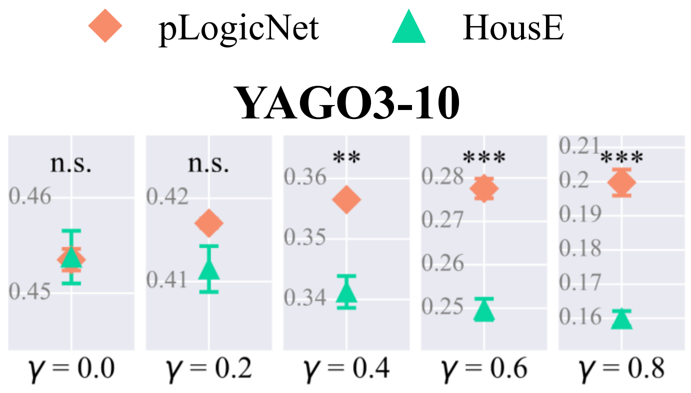


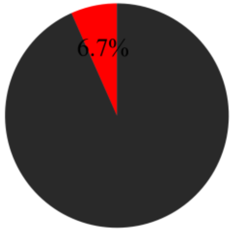


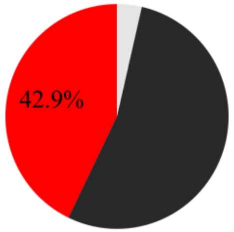


Rank

	ComplEx	RNNLogic
The Last King of Scotland, ./genre, docudrama	**39**	188
The Last King of Scotland, ./genre, thriller	**572**	798
The Last King of Scotland, ./genre, war film	**80**	751
Magic Mike, ./film_release_region, United Kingdom	20	**10**
Magic Mike, ./film_release_region, Finland	7	**2**





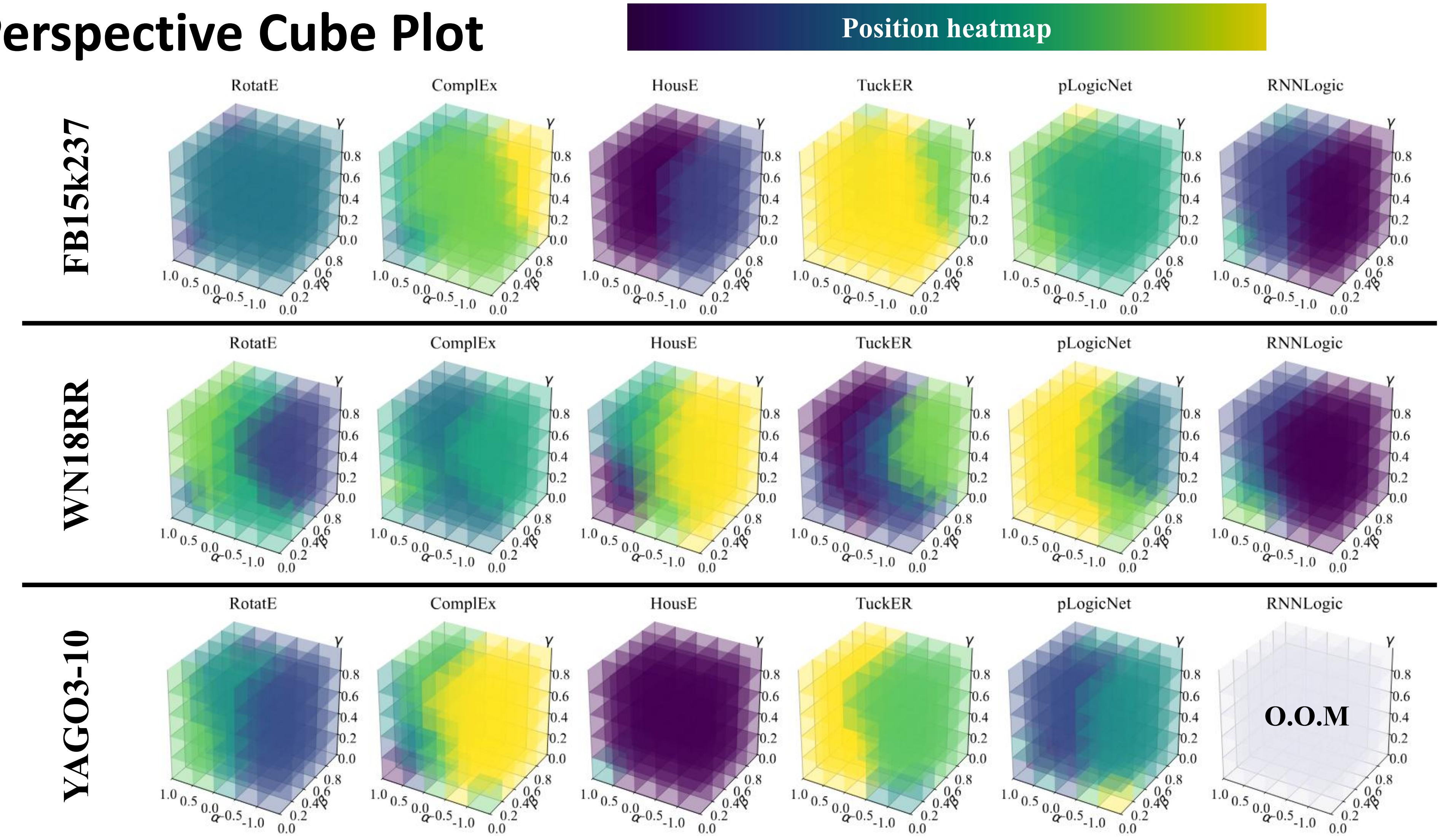


	pLogicNet	HousE
Bridgeport_Sound_Tigers, isLocatedIn, Fairfield_County,_Connecticut	**164**	2610
Massimo_Maccarone, playsFor, Italy_national_under-20_football_team	50	**5**

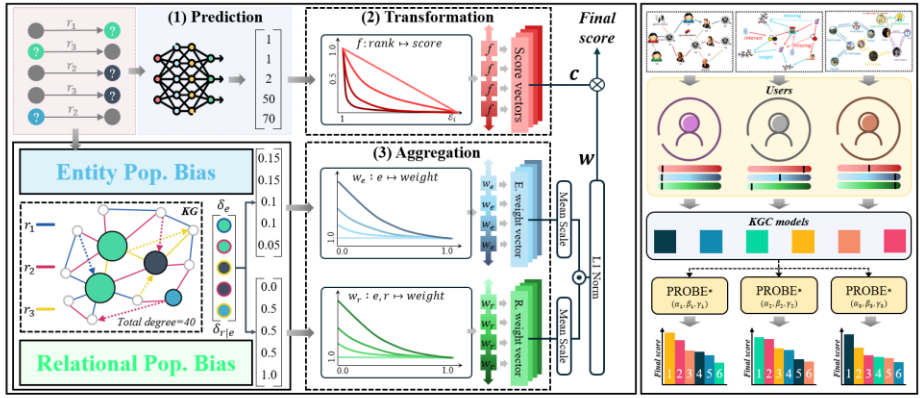
Progress bar

EXPERIMENTS

■ Tri-Perspective Cube Plot



Conclusion & Future Works



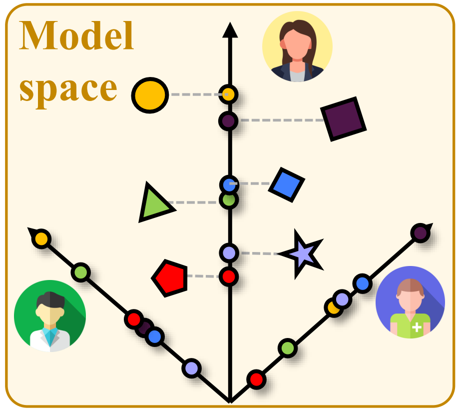
■ Motivation for Perspective-wise Metric

☐ Different objectives & use-cases of KGC models

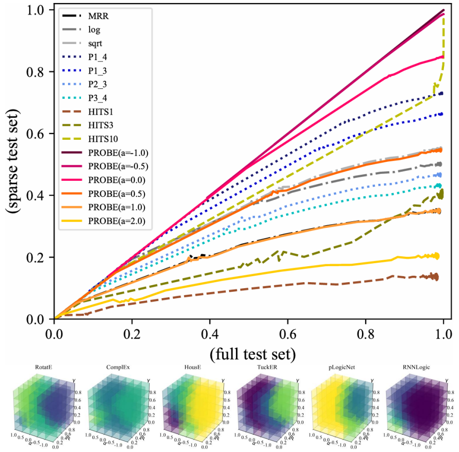
■ Generalized Framework (PROBE*)

☐ Predictive sharpness : degree of penalization

☐ Popularity bias robustness : degree of robustness on low popular entities / pairs



PROBE*



■ Comprehensive Analysis

☐ Theoretically analyze relationships between related works

■ Experiments & Case Studies

☐ Validated evaluation results of PROBE*

Conclusion & Future Works

■ Impact of Popularity Bias on the OW

- ☐ Synthesized family KG has near uniform degree distribution
- ☐ Investigating RA under more realistic KG (power law)

■ Deeper Analysis on RA

- ☐ A perspective constraint imposed to the KG itself
- ☐ Need theoretical justifications other than heuristic weight functions

■ Are There Method-Wise Strength or Weakness?

- ☐ Do some methods have advantages over these perspectives? If so, Why?

Q&A

Appendix

A. Proof of Lemma 1 and 2

Lemma 1. $Hits@1\ is\ equal\ to\ PROBE_{\alpha\to\infty}$

$$PROBE_{\alpha\to\infty} = \lim_{\alpha\to\infty} \frac{r^{-\alpha} - \mathcal{E}_i^{-\alpha}}{1 - \mathcal{E}_i^{-\alpha}}$$

$$= \begin{cases} 1 & \text{if } r = 1 \\ 0 & \text{if } r > 1 \end{cases}$$

Lemma 2. $PROBE_{\alpha=-1}\ is\ a\ chance\ adjusted\ metric$

Let's assume $r_i \sim Uniform(\{1, 2, ..., \mathcal{E}_i\})$ where $\mathbb{E}[r_i] = \frac{\mathcal{E}_i+1}{2}$

$$\mathbb{E}[PROBE_{\alpha=-1}] = \mathbb{E}[\frac{1}{n}\sum_{i=1}^{n}\frac{r_i - \mathcal{E}_i}{1 - \mathcal{E}_i}]$$

$$= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[\frac{r_i - \mathcal{E}_i}{1 - \mathcal{E}_i}]$$

$$= \frac{1}{n}\sum_{i=1}^{n}\frac{\mathbb{E}[r_i] - \mathcal{E}_i}{1 - \mathcal{E}_i}$$

$$= \frac{1}{n}\sum_{i=1}^{n}\frac{1 - \mathcal{E}_i}{2(1 - \mathcal{E}_i)}$$

$$= \frac{1}{2}.$$

Appendix

■ B. Proof of Lemma 3 (1)

Lemma 3. α *strictly decreases PROBE* $*$ expectation w.r.t model strength

We first discuss the bounds of variable k and $\hat{N}$ (absolute lower bound of rank). According to Corollary 4.1 from Yang et al. [32], the derivative of metric expectation w.r.t model strength is

$$\frac{\mathrm{d}\,\hat{\mathbb{E}}(\mathfrak{M})}{\mathrm{d}\ell} = \frac{1}{\ell\beta(N+1)}\,\mathbb{E}_{k\sim\mathcal{B}(N+1,\ell\beta)}\,g(k). \tag{5}$$

The random variant $\mathbf{m}$ is the number of missing answers from the number of full answers N which follows the binomial distribution $\mathbf{m} \sim \mathcal{B}(N,\beta)$. Thus, the number of filtered entity is $N-\mathbf{m}$ which results in $\hat{N} = N_e - (N-\mathbf{m})$. Here, $g(0) = 0$ by Yang et al. [32], which leads to the domain bound $1 \le k \le N+1$ rather than $0 \le k \le N+1$. By following the statement $N/N_e < 10\%$ *in almost all queries* [32], we can expand the domain bound to $1 \le k \le N+1 < \hat{N} \le N_e$.

The transform function and function $g_\alpha(k)$ is denoted as follows.

$$f^*_\alpha(k) := \frac{k^{-\alpha} - \hat{N}^{-\alpha}}{1 - \hat{N}^{-\alpha}} \qquad g_\alpha(k) := k f^*_\alpha(k), \qquad 1 \le k \le N+1.$$

Note that we don't consider $\alpha = 0$ since f^*_0 is not strictly defined.

Appendix

B. Proof of Lemma 3 (2)

Since both the denominator and the binomial distribution from 5 doesn't rely on α, the sign of the partial derivative solely relies on $g_\alpha(k)$. Thus we can reduce the question to *prove* $g_\alpha(k)$ *strictly decreases respect to* α.

$$\frac{\partial}{\partial\alpha}g_\alpha(k) = \frac{\partial}{\partial\alpha}kf_\alpha^*(k) \tag{6}$$

$$\propto \frac{\partial}{\partial\alpha}f_\alpha^*(k) \tag{7}$$

$$= \hat{N}^\alpha k^{-\alpha}(\hat{N}^\alpha - 1)(\ln\hat{N} - lnk) - \hat{N}^\alpha(\hat{N}^\alpha k^{-\alpha} - 1)ln\hat{N} \tag{8}$$

$$= \hat{N}^\alpha k^{-\alpha}((k^\alpha - 1)\ln\hat{N} + (1 - \hat{N}^\alpha)\ln k) \tag{9}$$

$$= \hat{N}^\alpha k^{-\alpha}F_\alpha(k). \tag{10}$$

Since $\hat{N}^\alpha k^{-\alpha} > 0$, we now have to prove that $F_\alpha(k) < 0$

$$\frac{\partial}{\partial\alpha}F_\alpha(k) = (k^\alpha - \hat{N}^\alpha)\ln k\ln\hat{N}.$$

Due to $k < \hat{N}$,

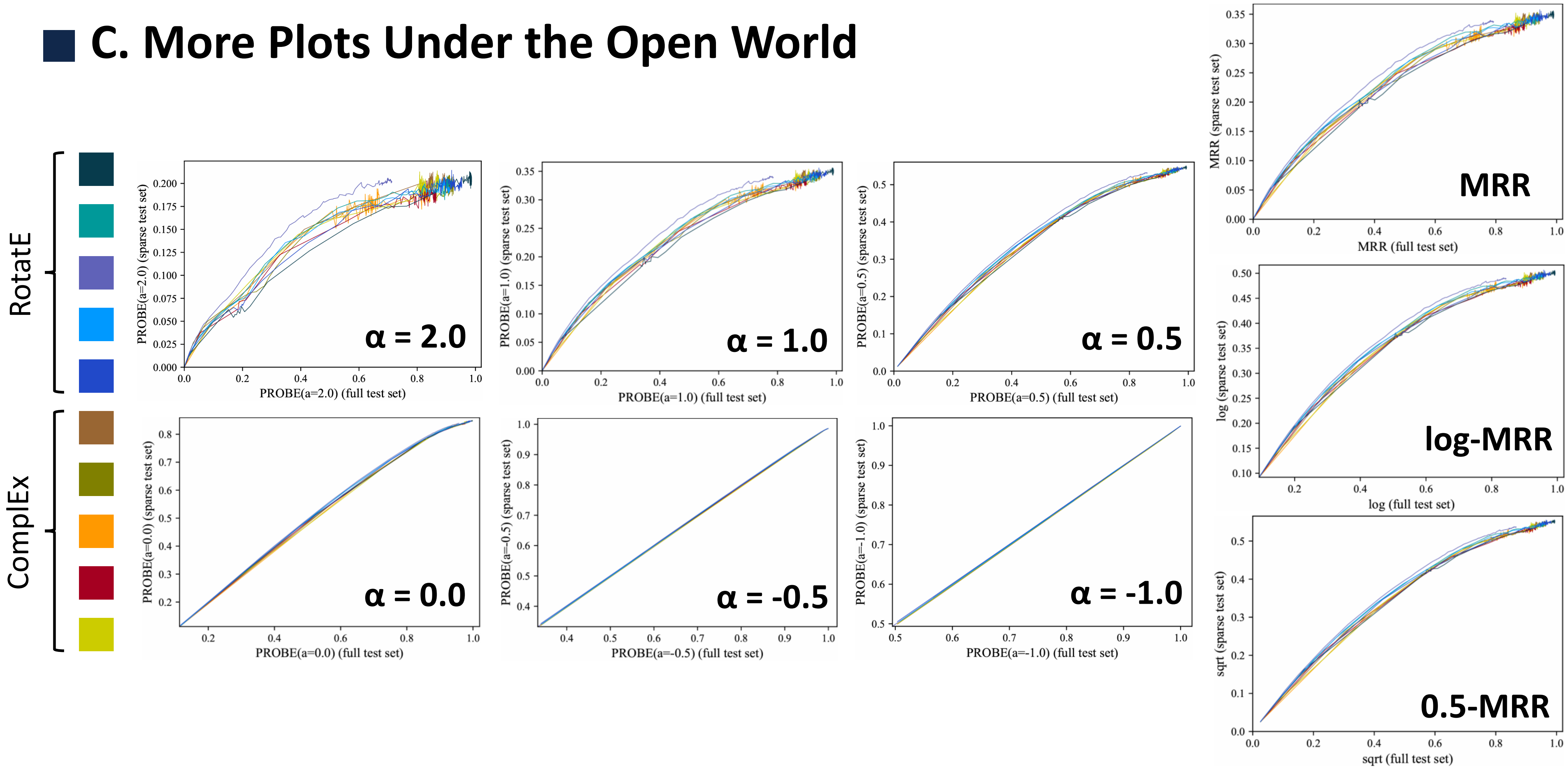
Case 1 $\alpha > 0$. $\frac{\partial}{\partial\alpha}F_\alpha(k) < 0$,

Case 2 $\alpha < 0$. $\frac{\partial}{\partial\alpha}F_\alpha(k) > 0$,

and since $\lim_{\alpha\to 0}F_\alpha(k) = 0$, we can conclude $F_\alpha(k) < 0$ for $\alpha \in \mathbb{R}\setminus\{0\}$ and thus, $\frac{\partial}{\partial\alpha}(\frac{\mathrm{d}\hat{\mathbb{E}}(PROBE_\alpha)}{\mathrm{d}\ell}) < 0$.

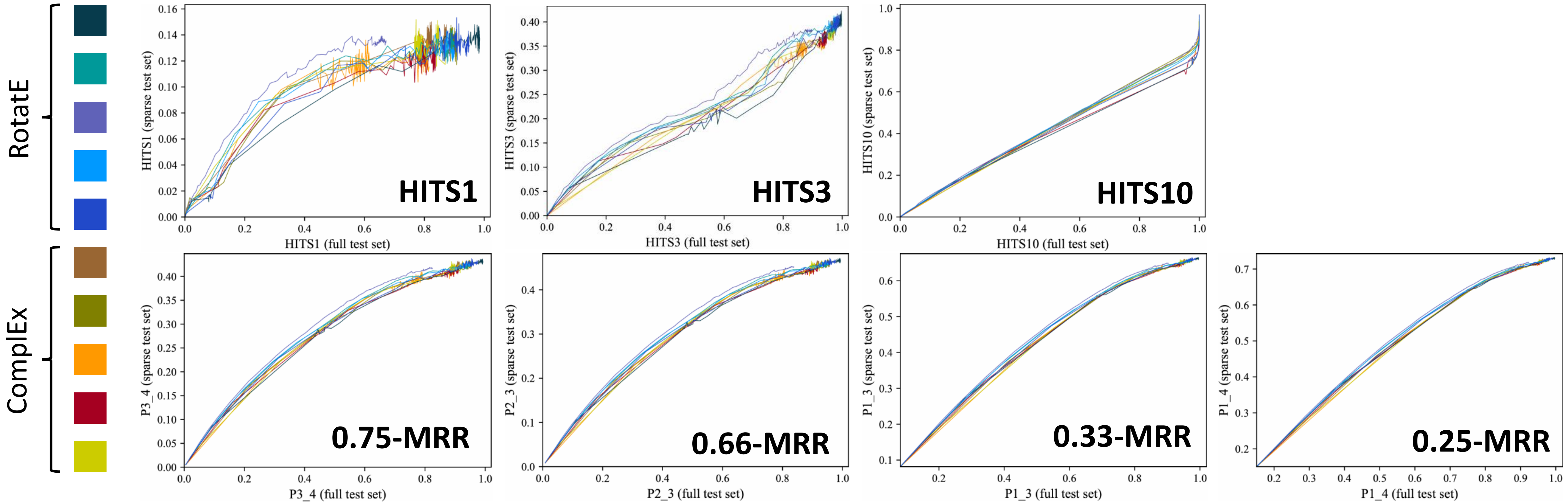
Appendix

■ C. More Plots Under the Open World



Appendix

■ C. More Plots Under the Open World



Appendix

■ D. More Tri-Perspective Plots